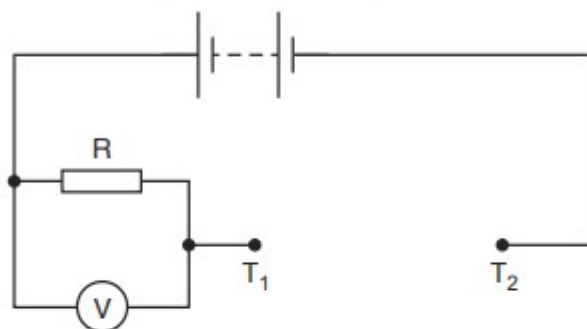


**1. O/N/23/22/Q7**

An electric circuit contains a battery, a fixed resistor  $R$ , a voltmeter and a thermistor.

Fig. 7.1 is an incomplete circuit diagram of the arrangement.



**Fig. 7.1**

The thermistor is connected between terminals  $T_1$  and  $T_2$ .

- (a) Complete Fig. 7.1 by drawing the circuit symbol for a thermistor connected between  $T_1$  and  $T_2$ . [1]

- (b) The resistance of  $R$  is  $2.5\Omega$ . The reading on the voltmeter is  $0.40\text{ V}$ .

Calculate:

- (i) the current in  $R$

current = ..... A [2]

- (ii) the power transferred in  $R$ .

power = ..... W [2]

- (c) The temperature of the thermistor increases.

Explain what happens to the reading on the voltmeter.

.....  
 .....  
 ..... [2]

## 2. O/N/23/21/Q6

Fig. 6.1 shows a circuit that contains two identical cells, each of e.m.f. 1.5V, that are connected in parallel.

The circuit also contains two ammeters, a thermistor T and two resistors R and S.

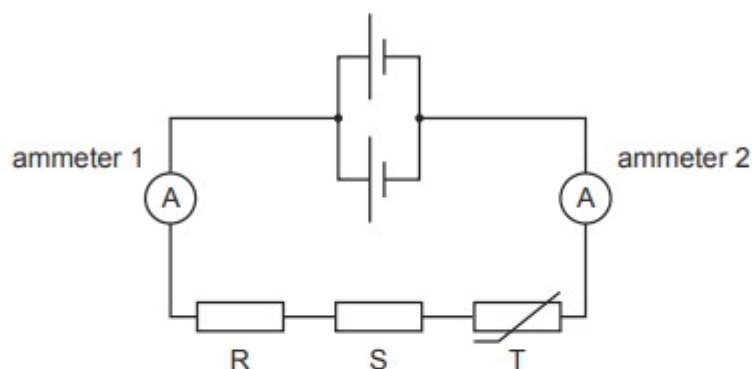


Fig. 6.1

The reading on ammeter 1 is 0.20A.

- (a) State why the reading on ammeter 2 is also 0.20A.

.....  
 ..... [1]

- (b) The resistance of R is  $1.5\Omega$  and the resistance of S is  $3.5\Omega$ .

- (i) Calculate the resistance of T.

resistance = .....  $\Omega$  [3]

- (ii) The temperature of T increases.

Explain why the potential difference (p.d.) across T decreases.

.....  
 .....  
 .....  
 .....  
 ..... [3]

3. M/J/23/22/Q6

Fig. 6.1 shows an electric circuit containing a filament lamp, a resistor R, a 12 V battery and five meters.

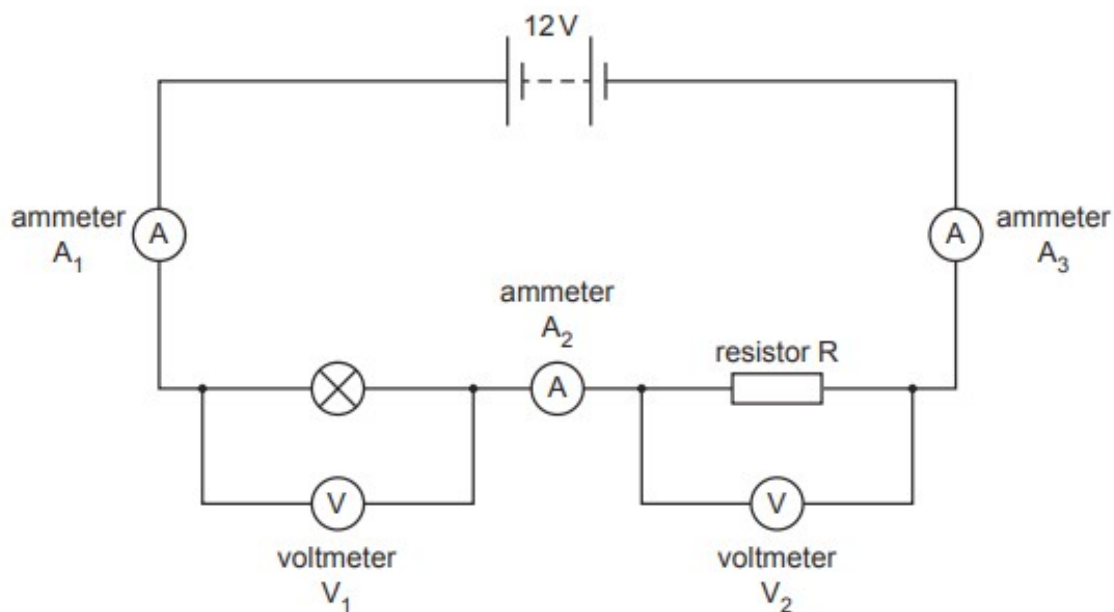


Fig. 6.1

- (a) (i) The reading on ammeter  $A_1$  is 0.25 A.

The reading on voltmeter  $V_1$  is 3.0 V.

Determine the readings on the other meters.

reading on ammeter  $A_2$  = ..... A

reading on ammeter  $A_3$  = ..... A

reading on voltmeter  $V_2$  = ..... V  
[2]

- (ii) Calculate the resistance of resistor R.

resistance of resistor R = .....  $\Omega$  [2]

(iii) The resistor obeys Ohm's law.

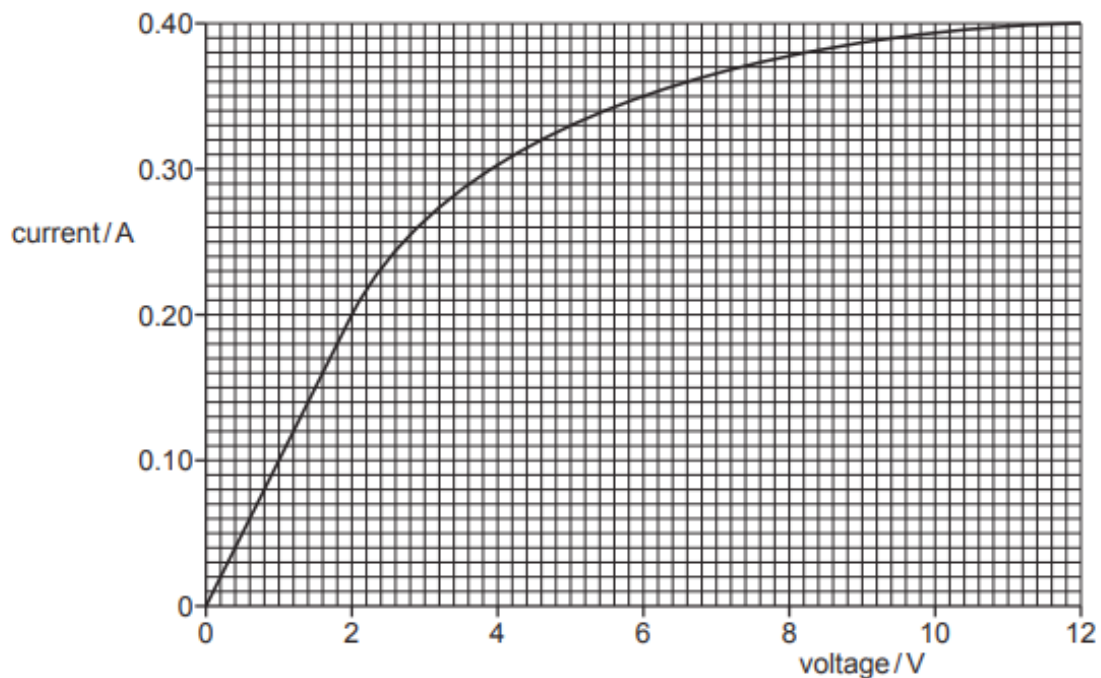
State Ohm's law.

.....

.....

..... [2]

(b) Fig. 6.2 shows the current–voltage graph for the filament lamp.



**Fig. 6.2**

The battery in Fig. 6.1 is replaced with a different battery which has a different e.m.f. (electromotive force).

The voltage across the lamp increases to 6.0 V.

Use data from the graph to determine the e.m.f. of the second battery.

Show your working.

e.m.f. = ..... V [3]

#### 4. M/J/23/21/Q3

Fig. 3.1 shows a circuit containing three resistors, a 12 V power supply and an ammeter.

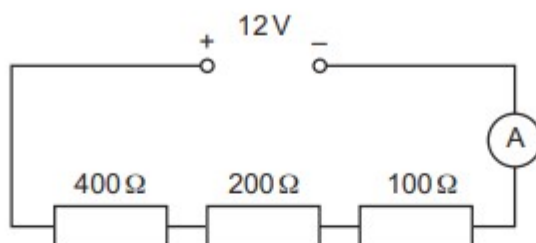


Fig. 3.1

- (a) Calculate the current in the 400  $\Omega$  resistor.

current = ..... A [2]

- (b) Calculate the power produced in the 400  $\Omega$  resistor.

power = ..... W [2]

- (c) A student uses all of the components shown in Fig. 3.1 in another circuit.

She connects them so that there is the largest possible current in each resistor.

The current in one of the resistors is larger than the current in the other two resistors. The student connects the ammeter into the circuit to measure the current in this resistor.

Draw the circuit diagram of the arrangement. Label each resistor with the value of its resistance.

5. O/N/22/22/Q5

The power supply in a circuit is made from several cells, each with the same electromotive force (e.m.f.). The cells are connected in parallel.

- (a) Explain what is meant by 'electromotive force'.

.....  
 .....  
 ..... [2]

- (b) State **one** advantage of using several cells in parallel rather than a single cell as the power supply.

.....  
 ..... [1]

- (c) The power supply is connected in series to a resistor of resistance  $4000\ \Omega$  and a thermistor. There is a voltmeter in parallel with the resistor.

Fig. 5.1 is the circuit diagram.

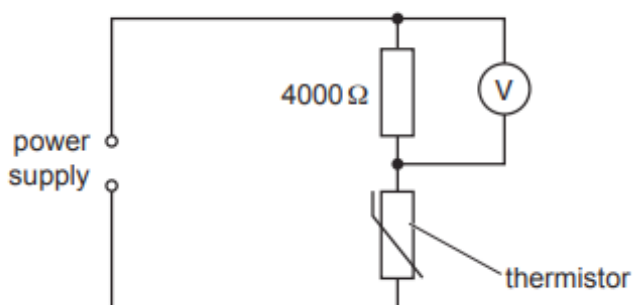


Fig. 5.1

- (i) The temperature of the thermistor increases.

Explain what happens to the reading on the voltmeter.

.....  
 .....  
 ..... [3]

- (ii) The e.m.f. of the power supply is 1.5 V.

Calculate the reading on the voltmeter when the resistance of the thermistor is  $8000\ \Omega$ .

reading = ..... [2]

**6. O/N/22/22/Q8**

An uncharged plastic rod is rubbed with an uncharged woollen cloth.

As a result, the rod becomes negatively charged.

- (a) Explain, in terms of particles, how the rod becomes charged and state what happens to the cloth.

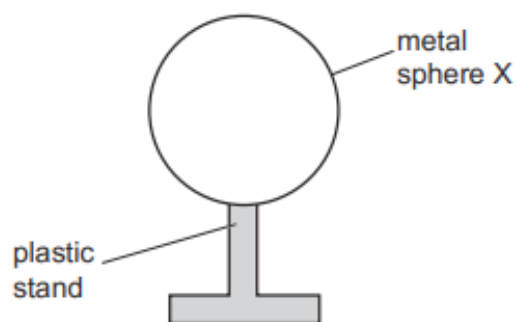
.....

.....

.....

..... [2]

- (b) Fig. 8.1 shows a metal sphere X on a plastic stand.



**Fig. 8.1**

Explain why metals are electrical conductors and why plastics are electrical insulators.

.....

.....

.....

..... [2]



- (d) A second metal sphere Y is negatively charged.

There is an electric field surrounding Y.

- (i) State what is meant by 'electric field'.

.....  
..... [1]

- (ii) Fig. 8.4 shows the negatively charged sphere Y. The negative charges are not shown.



**Fig. 8.4**

On Fig. 8.4, draw the pattern and mark the direction of the electric field lines surrounding sphere Y. [3]



7. M/J/22/22/Q5

(a) A student connects a battery, thermistor X and resistor Y in parallel, as shown in Fig. 5.1.

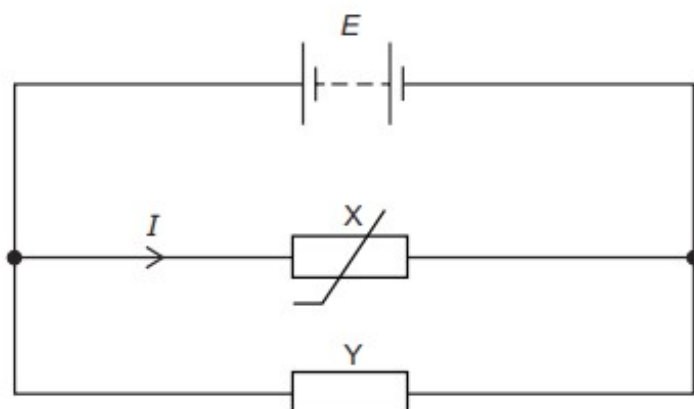


Fig. 5.1

- The electromotive force (e.m.f.) of the battery is  $E$ .
- The current in  $X$  is  $I$ .
- The resistance of  $Y$  is greater than the resistance of  $X$ .

Tick **two** boxes, one to show the correct statement about the potential difference across  $Y$  and the other to show the correct statement about the current in  $Y$ .

The potential difference across  $Y$ :

is less than  $E$

☐

is equal to  $E$

☐

is greater than  $E$ .

☐

The current in  $Y$ :

is less than  $I$

☐

is equal to  $I$

☐

is greater than  $I$ .

☐

[1]

- (b) The student connects thermistor X in series with the resistor Y and a battery of e.m.f. 6.0V, as shown in Fig. 5.2.

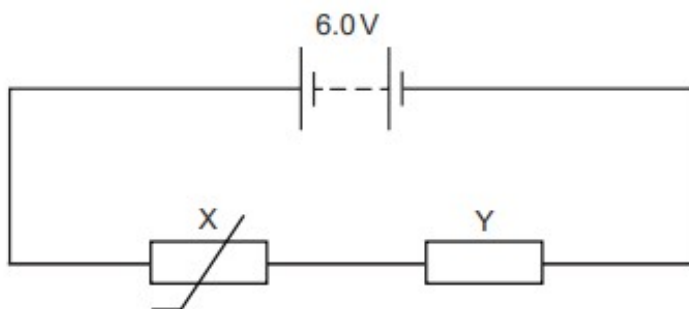


Fig. 5.2

In this circuit, at room temperature, the resistance of thermistor X is  $600\Omega$  and the current in thermistor X is 0.0020 A.

- (i) Calculate the power produced in the thermistor.

power = ..... [2]

- (ii) Calculate the resistance of Y.

resistance = ..... [3]

- (iii) The thermistor is cooled.

Explain why this causes the potential difference across Y to decrease.

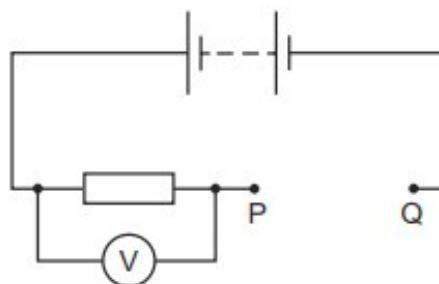
.....  
 .....  
 ..... [2]

[Total: 8]

**8. M/J/22/21/Q6**

- (a) A light-dependent resistor (LDR) is used to sense the amount of light in a room.

Fig. 6.1 shows part of the circuit used. The LDR is not shown.



**Fig. 6.1**

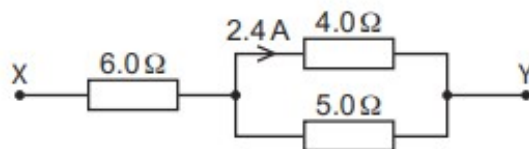
- (i) Complete Fig. 6.1 by drawing the circuit symbol for an LDR between points P and Q. [1]
- (ii) Explain how the voltmeter reading changes as the amount of light falling on the LDR increases.

.....

.....

..... [2]

- (b) Fig. 6.2 shows part of a circuit containing three resistors.



**Fig. 6.2**

The current in the  $4.0\ \Omega$  resistor is  $2.4\ \text{A}$ .

- (i) Show that the current in the  $5.0\ \Omega$  resistor is  $1.9\ \text{A}$ .

- (ii) Calculate the potential difference (p.d.) between points X and Y.

[2]

p.d. = ..... [2]

9. O/N/21/22/Q5

Fig. 5.1 shows a circuit that contains a direct current (d.c.) power supply, a light-dependent resistor (LDR), two fixed resistors, M and N, and two ammeters.

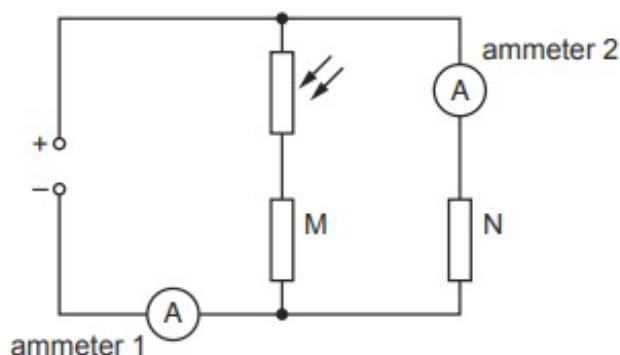


Fig. 5.1

- (a) The reading on ammeter 1 is 1.6 mA.

Calculate the time taken for a charge of 7.2 C to pass through ammeter 1.

time = ..... [2]

- (b) Some time later, the brightness of the light incident on the LDR decreases.

Explain what happens to:

- (i) the reading on ammeter 2

.....  
 ..... [1]

- (ii) the time taken for a charge of 7.2 C to pass through ammeter 1.

.....  
 .....  
 ..... [2]

[Total: 5]

10. O/N/21/21/Q6

A circuit contains a battery of electromotive force (e.m.f.) 4.5V, a  $28\Omega$  resistor and a variable resistor. Fig. 6.1 is the circuit diagram.

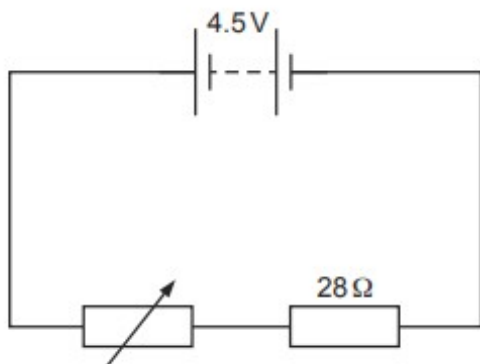


Fig. 6.1

- (a) The battery consists of three identical cells in series.

Determine the e.m.f. of each cell.

e.m.f. of each cell = ..... [1]

- (b) The potential difference (p.d.) across the variable resistor is 2.5V.

- (i) State what is meant by *potential difference*.

.....  
.....  
.....  
..... [2]

- (ii) Calculate the resistance of the variable resistor.

resistance = ..... [3]

[Total: 6]

**11. M/J/21/22/Q6**

The power input to a filament lamp when connected to a 12 V supply is 20 W.

- (a) State what is meant by *power*.

.....  
..... [1]

- (b) Calculate:

- (i) the current in the filament

current = ..... [2]

- (ii) the resistance of the filament.

resistance = ..... [1]

- (c) The filament in another lamp is made from thinner wire of the same length and made from the same material. Both lamps are connected to a 12 V supply.

State and explain how using thinner wire in the filament affects the power input to the lamp.

.....  
.....  
..... [2]

[Total: 6]

12. M/J/21/22/Q8

Fig. 8.1 shows a potential divider circuit containing two resistors  $R_1$  and  $R_2$ .

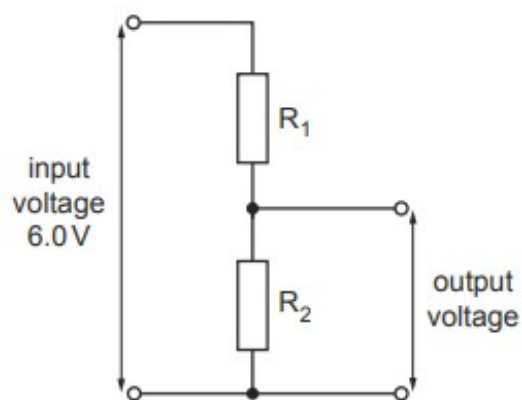


Fig. 8.1

- (a) There is an input voltage of 6.0 V.

Complete Table 8.1 to show the output voltage for different values of the resistances of the two resistors.

Table 8.1

| resistance of $R_1 / \Omega$ | resistance of $R_2 / \Omega$ | output voltage / V |
|------------------------------|------------------------------|--------------------|
| 8.0                          | 8.0                          | .....              |
| 800                          | 1200                         | .....              |

[3]



**(b)** In the circuit shown in Fig. 8.1, resistor  $R_1$  is replaced by a light dependent resistor (LDR).

**(i)** In the space below, draw the circuit symbol for an LDR.

[1]

**(ii)** The LDR is moved from darkness into daylight.

State what happens to the resistance of the LDR and the output voltage.

resistance .....

output voltage .....

[1]

**(iii)** A potential divider circuit can also be used to produce an output voltage that changes with temperature.

State the name of the component that is used instead of the LDR.

..... [1]

13. M/J/21/21/Q7

(a) Fig. 7.1 shows two circuit components.



Fig. 7.1

For each of these components state what causes an **increase** in its resistance.

thermistor .....

.....

LDR .....

.....

[2]

(b) Fig. 7.2 shows a battery connected to a diode, an ammeter and a resistor.

Fig. 7.3 shows the current–voltage graph for the diode.

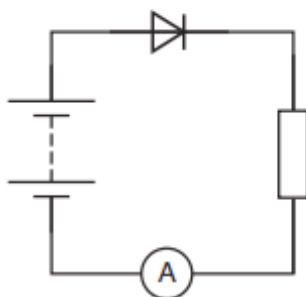


Fig. 7.2

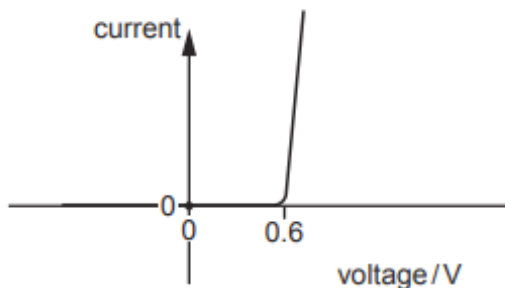


Fig. 7.3

The current in the resistor is 2.0 A and the potential difference across the resistor is 5.4 V.

(i) State what is meant by the *electromotive force* (e.m.f.) of the battery.

.....

..... [1]

(ii) Explain why the e.m.f. of the battery must be greater than 5.4 V.

.....

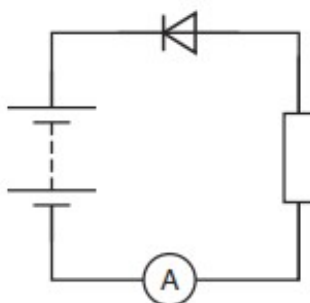
.....

..... [2]

- (iii) Calculate the electrical energy input to the resistor in 5.0 minutes.

energy input = ..... [2]

- (iv) The circuit is reconnected with the diode in the opposite direction, as shown in Fig. 7.4.



**Fig. 7.4**

Explain what is observed.

.....  
.....  
..... [1]

[Total: 8]

14. M/J/21/21/Q9

Fig. 9.1 is the current–voltage graph for a filament lamp.

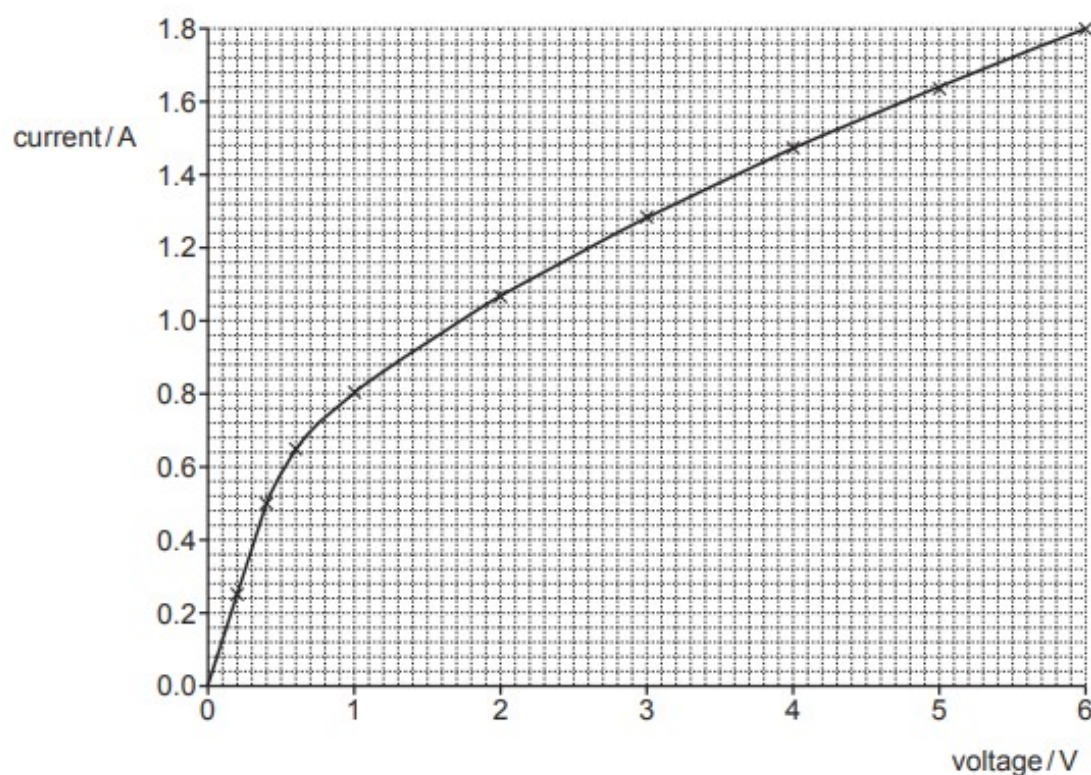


Fig. 9.1

- (a) Fig. 9.2 shows an incomplete circuit diagram of the circuit that a student uses to obtain the readings for the graph.

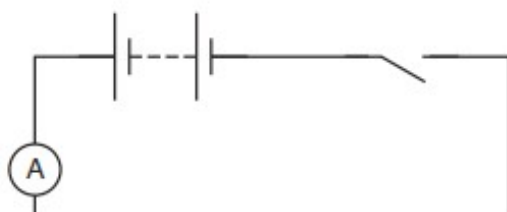


Fig. 9.2

Complete the circuit diagram using suitable circuit symbols for the lamp, a voltmeter and a variable resistor. [3]

- (b) The ammeter used by the student has ranges 0–10A, 0–1A, 0–100mA and 0–10mA.

Describe how the student uses the different ranges to obtain readings which allow all nine points to be plotted precisely on the graph while using the ammeter safely.

.....  
.....  
.....  
..... [2]

- (c) State how Fig. 9.1 shows that the lamp does **not** obey Ohm's law.

.....  
..... [1]

- (d) Using Fig. 9.1, determine the resistance of the lamp when:

- (i) the voltmeter reading is 0.40 V

resistance = .....

- (ii) the voltmeter reading is 6.0 V.

resistance = .....  
[3]

- (e) (i) Explain why the resistance of the lamp changes as the voltage changes.

.....  
.....  
..... [2]

- (ii) State how the resistance of a wire depends upon its length and how it depends on its cross-sectional area.

length .....  
.....  
cross-sectional area .....  
..... [2]

- (iii) The student obtains a long sample of the same type of wire as the filament in the lamp.

The wire he obtains has the same cross-sectional area as the filament.

He cuts a 1.0 m length of the wire and measures its resistance at room temperature.

He obtains the value  $16\ \Omega$ .

Using the appropriate resistance value from (d), estimate the length of the filament wire in the lamp.

length = ..... [2]

[Total: 15]

15. O/N/20/21/Q10

Fig. 10.1 shows a wire of length  $l$  and cross-sectional area  $A$ .

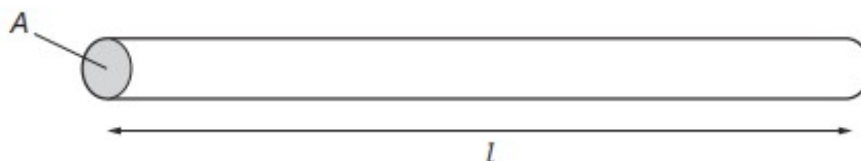


Fig. 10.1

(a) State how the resistance of the wire in Fig. 10.1 depends on:

(i)  $l$

..... [1]

(ii)  $A$ .

..... [1]

(b) The cross-sectional area of a piece of metal wire is  $7.5 \times 10^{-4} \text{ cm}^2$ . The resistance of a 1.0 m length of the same wire is  $6.4 \Omega$ .

The wire is made from metal W.

Fig. 10.2 shows a solid cube of side 1.0 cm. It is also made from metal W.

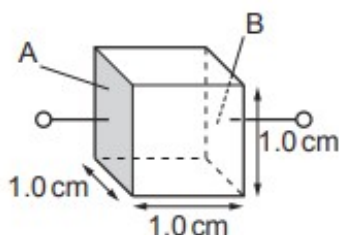


Fig. 10.2

Calculate the resistance between the two opposite faces A and B of the cube.

resistance = ..... [2]



- (c) The wire in part (b) is taped to a metre rule.

Fig. 10.3 shows that a 1.0 m length of the wire (resistance  $6.4\ \Omega$ ) is connected in series with a switch, a cell of electromotive force (e.m.f.) 1.2 V and a resistor of resistance  $9.6\ \Omega$ .

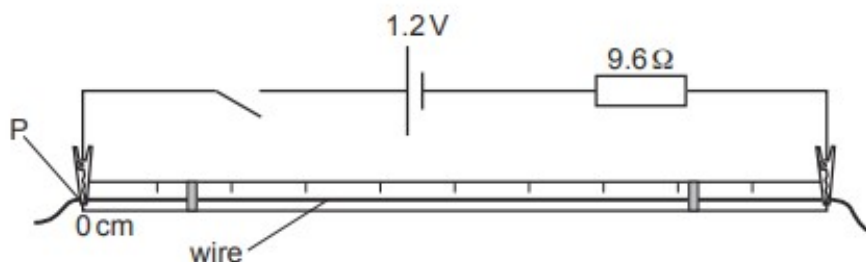


Fig. 10.3

The switch is closed.

- (i) Explain what is meant by *electromotive force (e.m.f.)*.

.....  
 .....  
 ..... [2]

- (ii) Calculate the potential difference (p.d.) across the 1.0 m length of the wire.

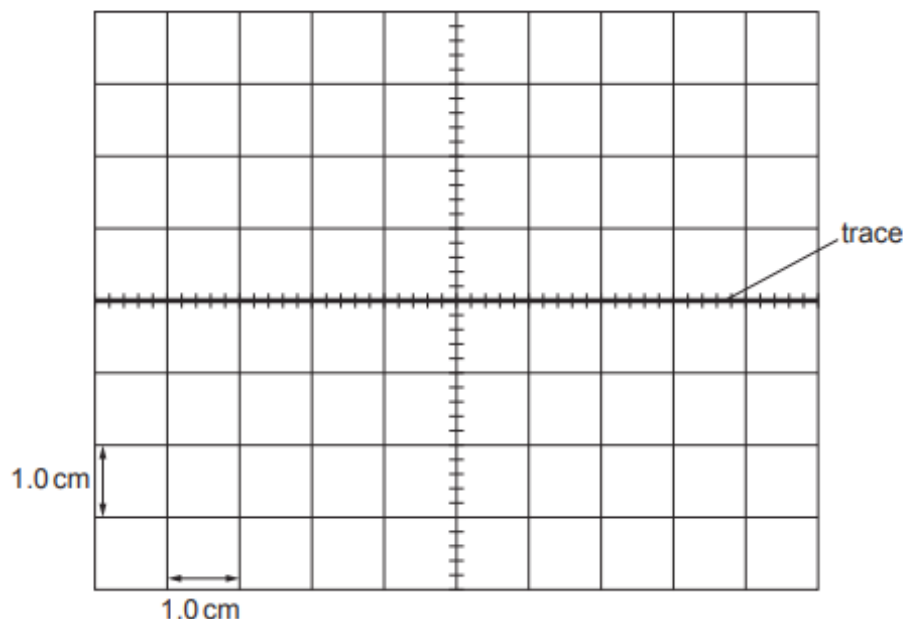
p.d. = ..... [3]

- (iii) One input terminal of an oscilloscope is connected to the wire at point P, the 0 cm mark of the metre rule.

The other terminal of the oscilloscope is connected to a sliding contact. Initially, this contact touches the wire at point P.

The Y-gain setting on the oscilloscope is  $0.20\text{ V/cm}$ .

Fig. 10.4 shows the screen of the oscilloscope with a horizontal trace across the middle of the screen.



**Fig. 10.4**

The sliding contact is slowly moved along the wire until it reaches the other end of the metre rule.

Describe and explain what happens to the trace on the screen.

.....

.....

.....

..... [3]

**(d)** A second, identical 1.2 V cell is connected in parallel with the cell in the circuit in Fig. 10.3.

**(i)** State **one** advantage of using two cells in parallel rather than a single cell.

.....

..... [1]

**(ii)** State and explain the effect on the trace in **(c)(iii)** of adding the second cell in parallel.

.....

.....

..... [2]

[Total: 15]

16. M/J/20/22/Q8

Fig. 8.1 shows the circuit diagram of a temperature gauge. It contains an ammeter, a thermistor, a fixed resistor  $R$  and a battery.

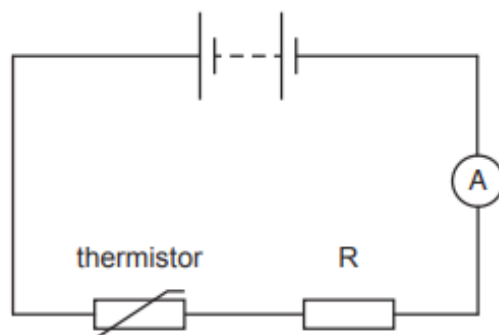


Fig. 8.1

The current is measured at different temperatures and a graph of the results is shown in Fig. 8.2.

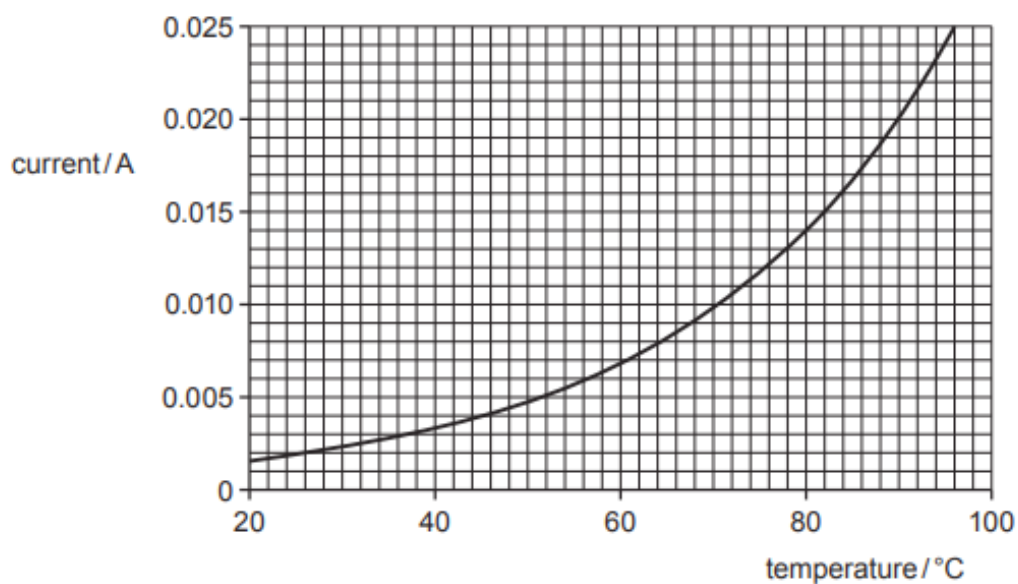


Fig. 8.2

- (a) (i) State how the resistance of the thermistor changes with temperature and explain how Fig. 8.2 shows this change.

.....

.....

..... [2]

- (ii) Resistance and current are used for the measurement of temperature.

State **one** other physical property that is used for the measurement of temperature.

.....  
..... [1]

- (b) A temperature scale involves the use of fixed points.

State what is meant by:

- (i) *the ice point*

.....  
..... [1]

- (ii) *the steam point.*

.....  
..... [1]

- (c) At  $80^{\circ}\text{C}$ , the potential difference (p.d.) across the thermistor is 3.6 V.

- (i) State what is meant by *potential difference*.

.....  
..... [1]

- (ii) Calculate the resistance of the thermistor when the temperature is  $80^{\circ}\text{C}$ .

resistance = ..... [3]

- (iii) The electromotive force (e.m.f.) of the battery is 25 V.

Calculate the resistance of the resistor R.

resistance = ..... [2]

17. M/J/20/21/Q8

Fig. 8.1 shows a lamp from a car. It contains two metal filaments.

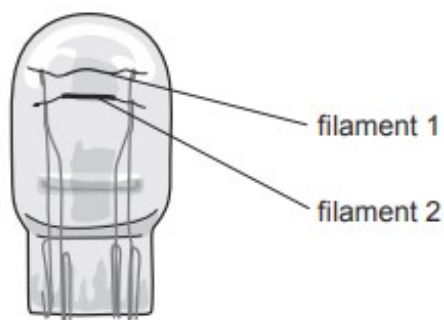


Fig. 8.1

- (a) (i) Complete the boxes to describe the transfer of energy that takes place when the lamp is switched on.

|              |   |                        |
|--------------|---|------------------------|
| ..... energy | → | ..... and ..... energy |
|--------------|---|------------------------|

[3]

- (ii) The efficiency of the metal filament lamp is less than 10%.

State what is meant by *efficiency*.

.....

..... [2]

- (b) The two filaments are usually connected in parallel to a car battery.

A student investigates what happens when the filaments are connected in series, rather than in parallel. He uses the same battery for the investigation.

State whether the current, the voltage across each filament and the total power produced *increases, decreases or stays the same* when the two filaments are connected in series.

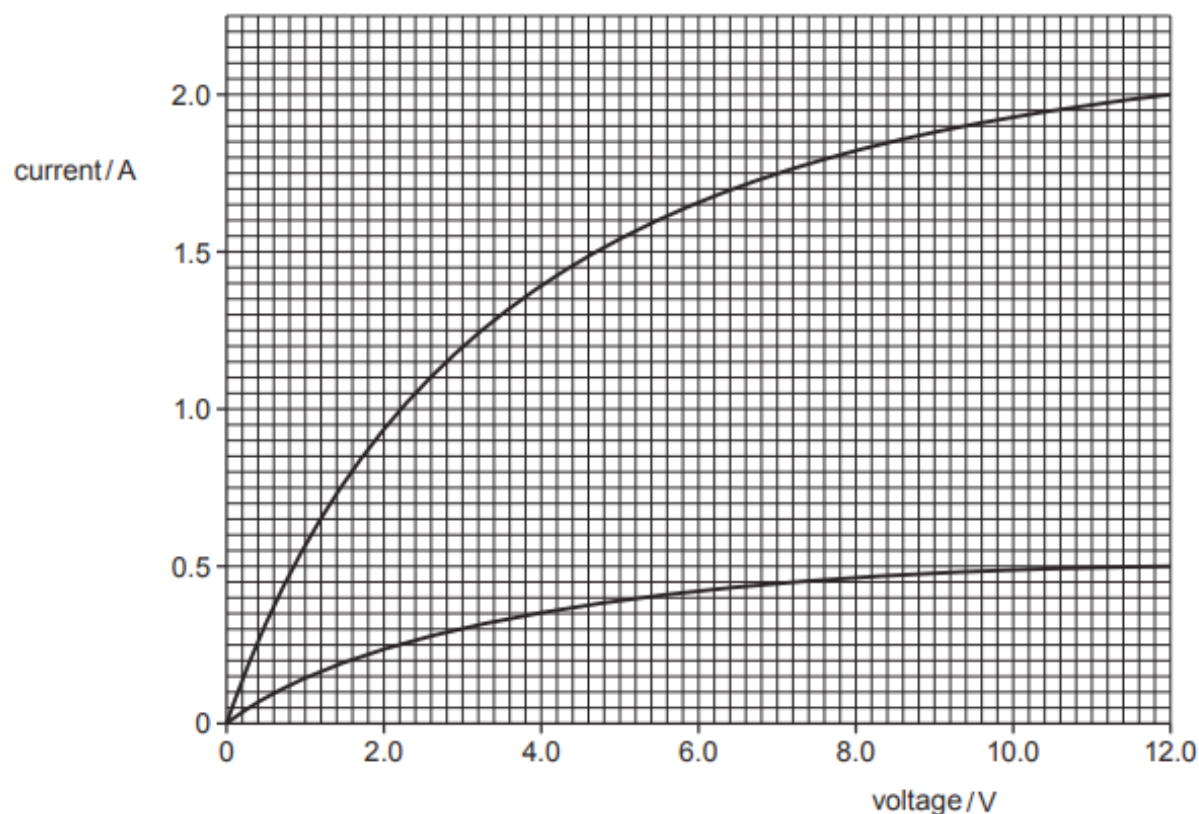
current .....

voltage .....

power .....

[2]

(c) Fig. 8.2 shows the current–voltage graph for the two filaments.



**Fig. 8.2**

- (i) Calculate the total resistance of the two filaments when they are connected in parallel to a voltage of 12 V.

resistance = ..... [3]

- (ii) The two filaments are made from the same type of metal and have the same length, when uncoiled. They both operate at the same temperature.

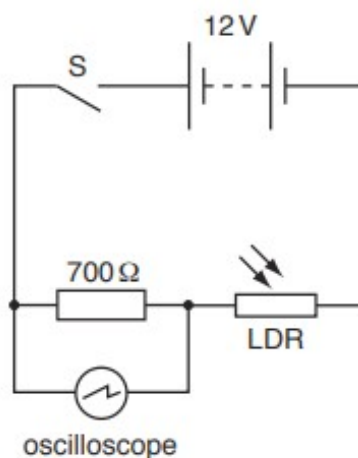
Suggest why one filament has a resistance that is greater than that of the other filament.

.....

..... [1]

**18. O/N/19/22/Q6**

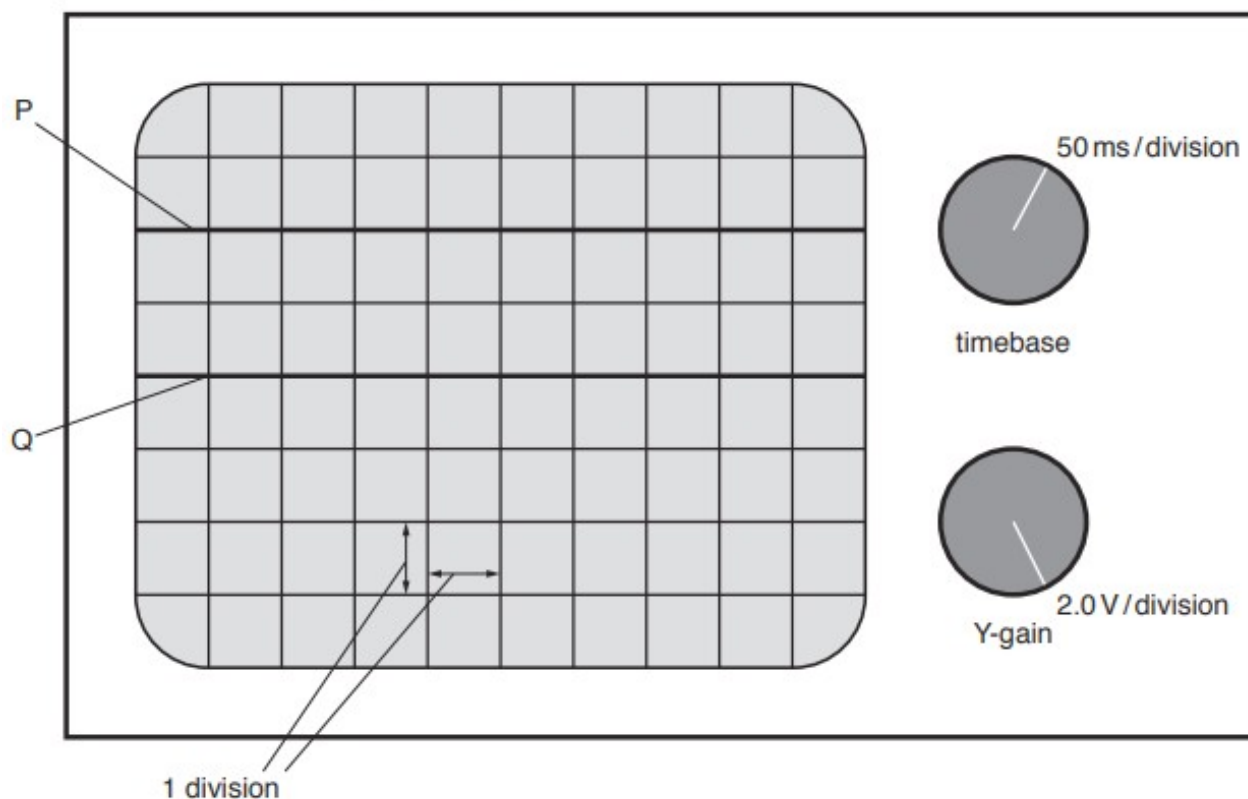
An electric circuit contains a  $700\ \Omega$  resistor and a light-dependent resistor (LDR). Fig. 6.1 is the circuit diagram.



**Fig. 6.1**

The electromotive force (e.m.f.) of the battery is 12 V.

An oscilloscope is connected across the fixed resistor. Fig. 6.2 shows the oscilloscope, including the settings of the timebase and the Y-gain controls. Line Q shows the position of the trace on the oscilloscope when the switch S is open.



**Fig. 6.2**



The switch S is closed and the trace on the oscilloscope moves to the position shown by line P in Fig. 6.2.

- (a) (i) Determine the potential difference (p.d.) across the  $700\,\Omega$  resistor.

p.d. = ..... [1]

- (ii) Determine the resistance of the LDR.

resistance = ..... [3]

- (b) The intensity of the light incident on the LDR gradually increases.

State and explain how the trace on the oscilloscope screen moves.

.....  
.....  
.....  
..... [3]

[Total: 7]

**19. O/N/19/21/Q6**

A 60 W filament lamp is powered by the 240 V mains electricity supply.

The lamp is switched on.

**(a)** Calculate:

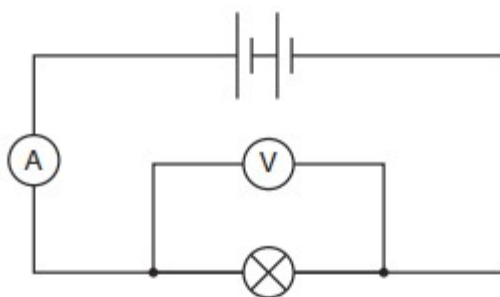
**(i)** the current in the lamp

current = ..... [2]

**(ii)** the resistance of the lamp.

resistance = ..... [2]

**(b)** The lamp is now unplugged from the mains supply and connected in series with two 1.5 V cells. Fig. 6.1 is the circuit diagram.



**Fig. 6.1**

**(i)** State the reading on the voltmeter.

reading = ..... [1]

**(ii)** State and explain how the resistance of the lamp now differs from the value calculated in **(a)(ii)**.

.....  
 .....  
 ..... [2]

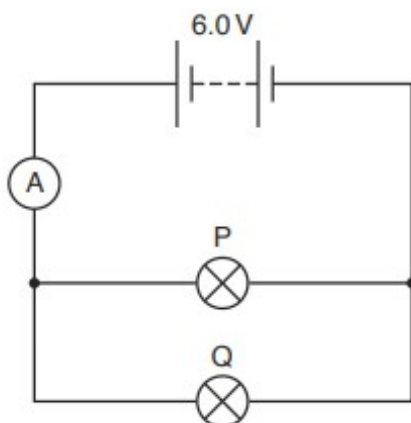
**(iii)** When a filament lamp blows, it very often does so immediately after being switched on.

Suggest why this is the case.

.....  
 ..... [1]

**20. M/J/19/22/Q6**

Two lamps, P and Q, are connected to a battery of electromotive force (e.m.f.) 6.0V and an ammeter, as shown in Fig. 6.1.



**Fig. 6.1**

Lamp P has a resistance of  $15\Omega$ . The ammeter reading is 0.65A.

**(a)** Calculate the current in lamp P.

current = ..... [2]

**(b)** Calculate the resistance of lamp Q.

resistance = ..... [2]

**(c)** The two lamps are now connected in series to the ammeter and the same battery.

**(i)** In the space below draw the circuit diagram.

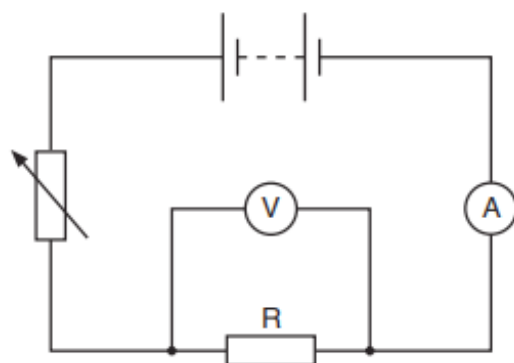
[1]

**(ii)** Explain why the ammeter reading is less than 0.65A when the lamps are connected in series.

.....  
 ..... [1]

**21. M/J/19/21/Q5**

A student uses the circuit shown in Fig. 5.1 to investigate a resistor R.



**Fig. 5.1**

- (a) Describe how the student uses the apparatus in Fig. 5.1 to obtain a range of ammeter and voltmeter readings.

.....  
 ..... [1]

- (b) The readings obtained are shown in Table 5.1.

**Table 5.1**

| voltmeter reading / V | ammeter reading / mA |
|-----------------------|----------------------|
| 7.6                   | 320                  |
| 5.2                   | 220                  |
| 2.4                   | 100                  |

- (i) State Ohm's law.

.....  
 .....  
 ..... [2]

- (ii) Using the results in Table 5.1, show that the resistor obeys Ohm's law.

.....  
 .....  
 .....  
 ..... [1]

(iii) The student has available an ammeter with five ranges marked:

- 0 – 10A
- 0 – 200mA
- 0 – 20mA
- 0 – 2mA
- 0 – 0.2mA

Suggest how the student makes best use of the different ranges during the investigation.

.....

.....

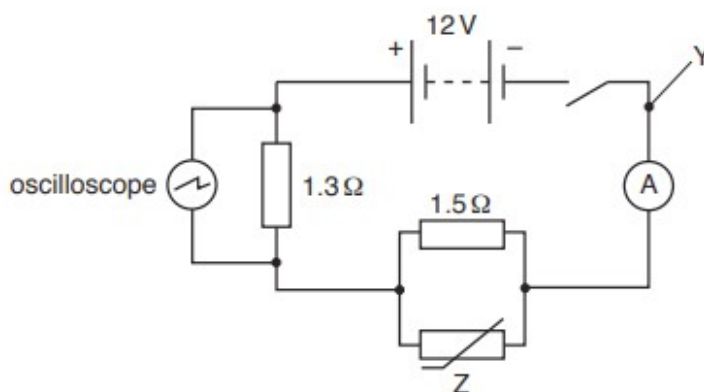
.....

..... [1]

[Total: 5]

**22. O/N/18/22/Q10**

A student sets up the circuit shown in Fig. 10.1 in a laboratory at room temperature.



**Fig. 10.1**

The electromotive force (e.m.f.) of the battery is 12 V.

The switch is closed.

- (a) The connecting wires in the circuit are made from copper covered by insulating plastic.

State the name of the particles that flow in the copper and state the direction in which they are flowing at point Y in the circuit.

.....  
 .....  
 ..... [2]

- (b) At room temperature, the resistance of component Z is  $6.0\ \Omega$ .

- (i) State the name of component Z.

..... [1]

- (ii) Calculate the resistance of the whole circuit.

resistance = ..... [3]

- (iii) Calculate the current measured by the ammeter.

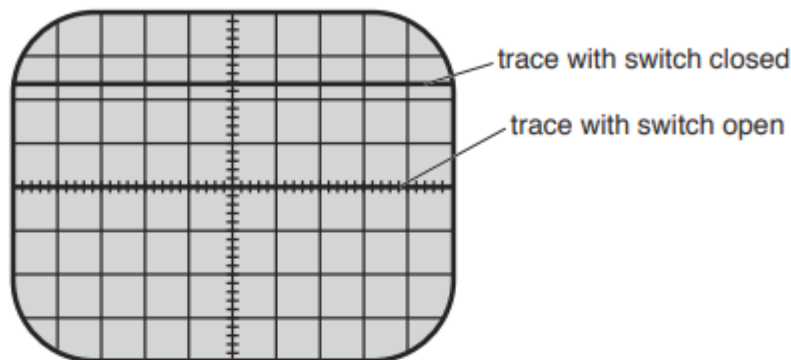
current = ..... [2]

- (iv) The current in the ammeter is  $I_A$ , the current in the  $1.5\Omega$  resistor is  $I_R$  and the current in component Z is  $I_Z$ .

Write down an equation that relates  $I_A$ ,  $I_R$  and  $I_Z$ .

..... [1]

- (c) Fig. 10.2 shows the screen of the oscilloscope.



**Fig. 10.2**

Before the switch is closed, the trace is a horizontal line across the middle of the screen, as shown in Fig. 10.2. When the switch is closed, the trace remains horizontal and moves up the screen.

Component Z is heated.

State and explain what is observed on the oscilloscope screen as the temperature of Z increases.

.....  
 .....  
 .....  
 .....  
 .....  
 ..... [4]

- (d) The 12 V battery is made from cells of electromotive force (e.m.f.) 1.5 V which are all in series.

- (i) Calculate the number of 1.5 V cells that make up the battery.

number = ..... [1]

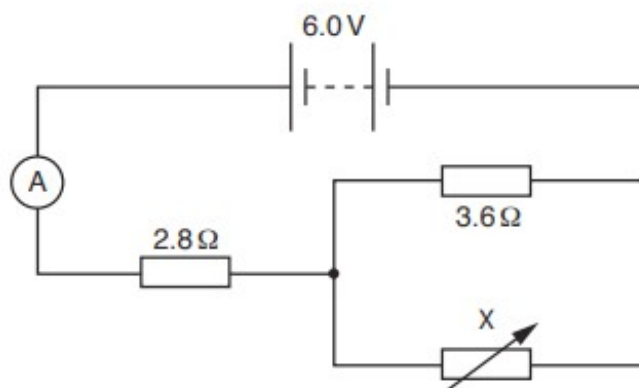
- (ii) State the e.m.f. of a battery made by connecting all of these cells in parallel.

e.m.f. = ..... [1]



**23. O/N/18/21/Q7**

A student sets up the circuit shown in Fig. 7.1.



**Fig. 7.1**

The electromotive force (e.m.f.) of the battery is 6.0 V.

- (a)** The resistance of the variable resistor X is set to  $1.8\Omega$ .

Determine

- (i)** the total resistance of the circuit,

resistance = ..... [3]

- (ii)** the current measured by the ammeter.

current = ..... [2]

- (b) (i)** State what is meant by the potential difference (p.d.) across a component in a circuit.

.....  
 ..... [1]

- (ii)** Add to Fig. 7.1 the symbol for a component that is measuring the p.d. across the  $2.8\Omega$  resistor. [1]

(iii) The resistance of X is increased.

Explain why the p.d. across the  $2.8\ \Omega$  resistor decreases.

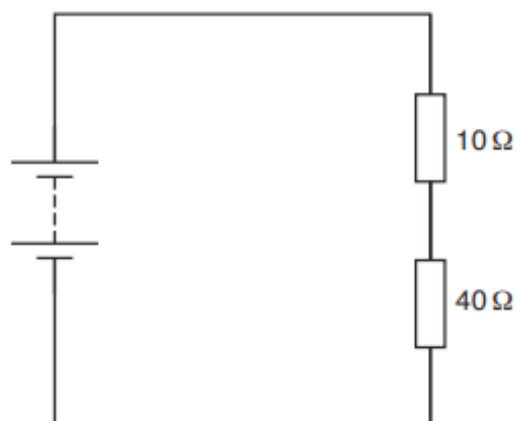
.....

.....

..... [2]

**24. M/J/18/22/Q10**

A student connects a battery to two resistors. The circuit diagram is shown in Fig. 10.1.



**Fig. 10.1**

The potential difference (p.d.) across the  $40\ \Omega$  resistor is  $9.6\ \text{V}$ .

- (a) State what is meant by the *potential difference* across a resistor.

.....  
 .....[2]

- (b) (i) Calculate the current in the  $40\ \Omega$  resistor.

current = .....[2]

- (ii) Calculate the electromotive force (e.m.f.) of the battery.

e.m.f. = .....[2]

- (c) The student has three different voltmeters to measure the p.d. across the  $40\ \Omega$  resistor. These are labelled  $0\text{--}2\ \text{V}$ ,  $0\text{--}20\ \text{V}$  and  $0\text{--}200\ \text{V}$ . Each has a pointer that shows the p.d. on a scale.

State and explain which of the three voltmeters is best to measure this p.d.

.....  
 .....  
 .....  
 .....[2]

- (d) (i) Calculate the power  $P$  produced in the  $10\Omega$  resistor.

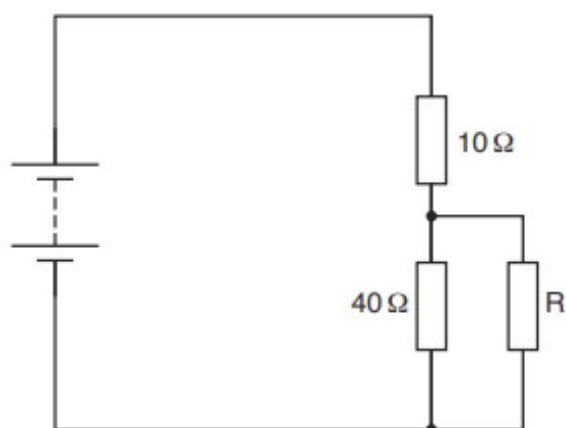
$$P = \dots\dots\dots [2]$$

- (ii) The student has available two  $10\Omega$  resistors, with power ratings of  $\frac{1}{2}P$  and  $2P$ .

Suggest why a resistor with a power rating of  $\frac{1}{2}P$  is not suitable for the circuit in Fig. 10.1.

.....  
 ..... [1]

- (e) The student adds a resistor  $R$  to the circuit, to make the circuit shown in Fig. 10.2.



**Fig. 10.2**

Complete the table in Fig. 10.3 to show what happens as resistor  $R$  is connected.

You should state whether each quantity increases, decreases or stays the same and give a brief explanation of why any change occurs. Calculations are not required.

| quantity                           | increases,<br>decreases or<br>stays the same<br>as resistor $R$ is<br>added | brief explanation of why the change occurs |
|------------------------------------|-----------------------------------------------------------------------------|--------------------------------------------|
| current in $10\Omega$<br>resistor  | increases                                                                   |                                            |
| p.d. across $10\Omega$<br>resistor |                                                                             |                                            |
| p.d. across $40\Omega$<br>resistor |                                                                             |                                            |

[4]

**Fig. 10.3**

**25. M/J/18/21/Q5**

A student has three resistors; each resistor is labelled  $3.9\Omega$ , 3 W.

She checks the value of the resistance of **one** of the resistors.

- (a) In the space below, draw a circuit diagram of a circuit that she can use to find this resistance. Include a battery, a variable resistor, a voltmeter and an ammeter in the circuit.

[2]

- (b) When the circuit is turned on, the potential difference (p.d.) across the resistor is 4.2 V and the current in the resistor is 1.2 A.

- (i) Calculate the resistance of the resistor.

resistance = .....[2]

- (ii) Shortly after the circuit is turned on, the resistor overheats and gives off smoke.

Explain why it overheats. Include a calculation in your answer.

.....  
.....  
.....  
.....  
.....[2]

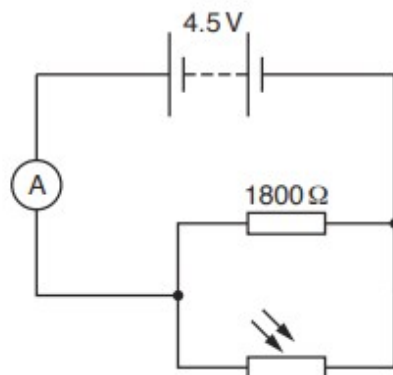
- (iii) The other two resistors are also labelled  $3.9\Omega$ , 3 W. The first resistor is now removed and the student connects these other two resistors in series to the p.d. of 4.2 V.

State and explain whether these resistors overheat.

.....  
.....  
.....  
.....[1]

**26. O/N/17/22/Q7**

A 4.5 V battery is connected in a circuit with an ammeter, a light-dependent resistor (LDR) and a  $1800\ \Omega$  fixed resistor. Fig. 7.1 is the circuit diagram.



**Fig. 7.1**

- (a)** The resistance of the LDR is  $9000\ \Omega$ .

Calculate

- (i)** the total resistance of the circuit,

resistance = .....[2]

- (ii)** the reading on the ammeter.

reading = .....[2]

- (b)** A very bright lamp is switched on and the light intensity on the LDR increases.

State and explain what happens to

- (i)** the current in the LDR,

.....  
 .....[1]

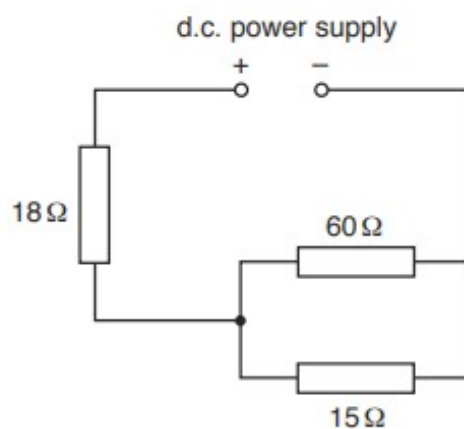
- (ii)** the current in the  $1800\ \Omega$  resistor.

.....  
 .....[1]

**27. O/N/17/21/Q11(b)**

**(b)** A circuit contains a 7.5 V d.c. power supply and three resistors.

Fig. 11.2 is the circuit diagram.



**Fig. 11.2**

**(i)** Determine the total resistance of the circuit.

resistance = ..... [3]

**(ii)** Calculate the current in the power supply.

current = ..... [2]

**(iii)** There is a different current in each of the three resistors in the circuit in Fig. 11.2.

Underline the resistor in which the current is

1. the largest,

**15  $\Omega$  resistor**

**18  $\Omega$  resistor**

**60  $\Omega$  resistor**

2. the smallest.

**15  $\Omega$  resistor**

**18  $\Omega$  resistor**

**60  $\Omega$  resistor**

[1]



- (iv) The 7.5 V d.c. power supply is made from five 1.5 V cells.

In the space below, draw a diagram that shows how the cells are arranged.

[1]

- (v) The  $18\ \Omega$  resistor is made of metal.

State and explain what happens to the current in this resistor as its temperature increases.

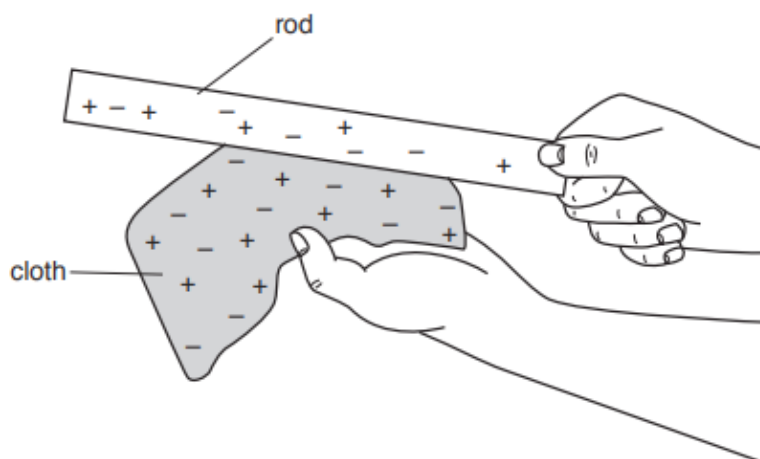
.....

.....

..... [1]

**28. M/J/17/22/Q8**

Fig. 8.1 shows charges on a rod and on a cloth.



**Fig. 8.1**

- (a) State how Fig. 8.1 shows that the rod and the cloth are uncharged.

[1]

- (b)** The plastic rod becomes negatively charged when rubbed by the cloth.

- (i) Describe what happens to the charges on the rod and on the cloth during the charging process.

.....

.....

.....

.....

[2]

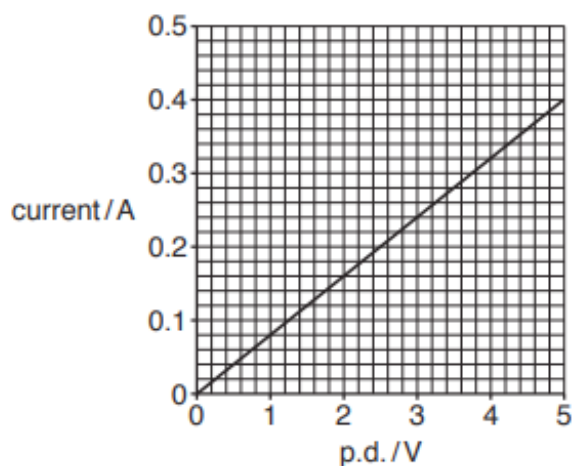
- (ii) Describe a simple experiment to show that the plastic rod is charged.

You may draw a diagram if you wish.

[2]

**29. M/J/17/22/Q10**

- (a) Measurements are made of the current in a metal wire and the potential difference (p.d.) across it. Fig. 10.1 shows a graph of the current against the p.d. for the wire.



**Fig. 10.1**

- (i) State the relationship between the current and the p.d. for the metal wire.

.....  
 .....[1]

- (ii) State how the graph shows that the temperature of the wire does not change in the experiment.

.....  
 .....[1]

- (iii) Another wire of the same material has the same length as the original wire but has only half the cross-sectional area.

1. Compare the resistance of the new wire and the original wire.

.....  
 .....[1]

2. On Fig. 10.1, draw the graph for the new wire. [1]

- (b) Three resistors are connected to a d.c. power supply, as shown in Fig. 10.2. The electromotive force (e.m.f.) of the power supply is 6.0 V.

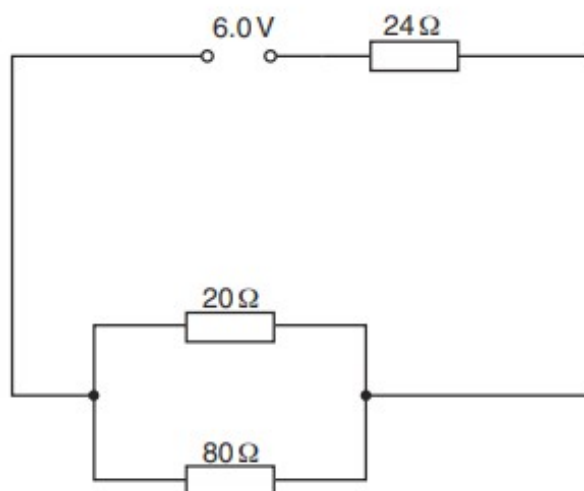


Fig. 10.2

(i) Calculate

1. the total resistance of the circuit,

resistance = .....[2]

2. the current supplied by the power supply,

current = .....[2]

3. the p.d. across the 20 Ω resistor.

p.d. = .....[2]

- (ii) The d.c. power supply is a battery that contains a number of cells. Each cell has an electromotive force (e.m.f.) of 1.5 V.

1. State what is meant by the *electromotive force (e.m.f.)* of a cell.

.....  
.....  
.....[1]

2. In the space below, draw a diagram to show how four of these cells are connected to make a battery of e.m.f. 6.0 V. Mark the positive terminal of the battery.

[2]

3. In another arrangement to produce a battery of e.m.f. 6.0 V, eight cells of e.m.f. 1.5 V are connected. In this arrangement, some cells are connected in series and some in parallel.

In the space below, draw a diagram to show how the eight cells are connected to make the battery.

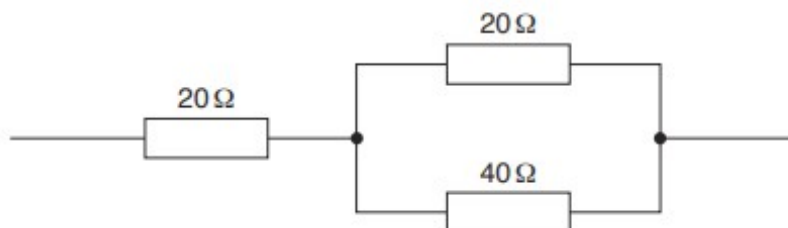
[1]

- (c) State one advantage of using a battery made from cells in parallel rather than a single cell.

.....  
.....[1]

**30. M/J/17/21/Q5**

Fig. 5.1 shows an arrangement of three resistors.



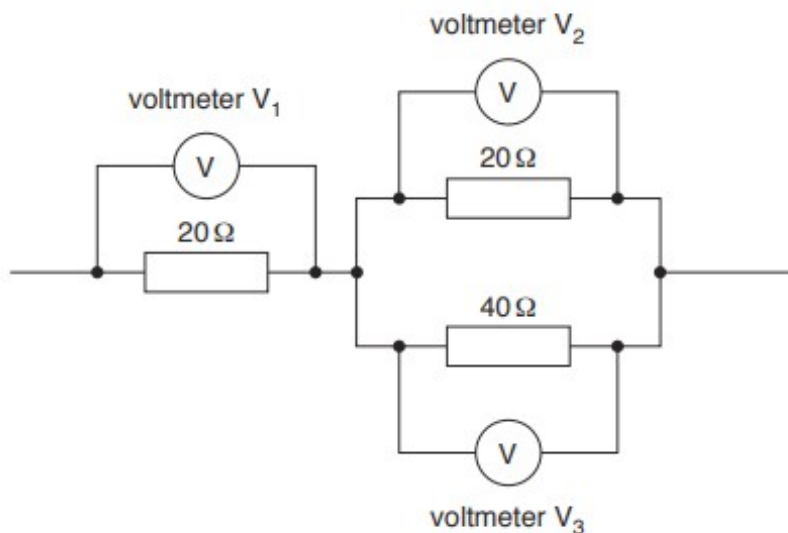
**Fig. 5.1**

- (a) Calculate the total resistance of this arrangement.

resistance = .....[3]

- (b) The arrangement is connected to a d.c. power supply.

Voltmeters are placed across the resistors, as shown in Fig. 5.2. There is a current in each resistor.



**Fig. 5.2**

Compare the readings on the three voltmeters. There is no need for any calculation.

.....  
 .....  
 .....[2]

**31. M/J/17/21/Q11**

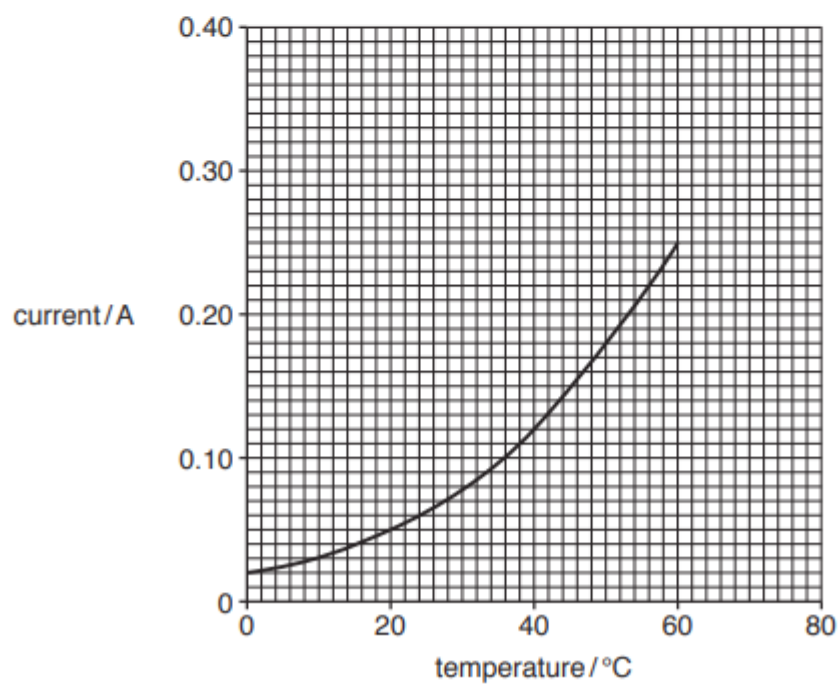
A student connects a thermistor in series with a battery and an ammeter. He uses the resistance of the thermistor to determine temperature.

- (a) In the space below, draw a circuit diagram showing the thermistor, the battery and the ammeter.

[3]

- (b) The student places the thermistor in a water bath at various temperatures and records the reading on the ammeter.

Fig. 11.1 shows how the current varies with temperature.



**Fig. 11.1**



The potential difference (p.d.) across the thermistor **remains constant**.

The resistance of the thermistor is  $240\ \Omega$  at a temperature of  $20^\circ\text{C}$ .

(i) Using Fig. 11.1, calculate

1. the potential difference (p.d.) across the thermistor,

p.d. = .....[3]

2. the resistance of the thermistor at a temperature of  $40^\circ\text{C}$ .

resistance = .....[2]

- (ii) The resistance of the thermistor is  $240\ \Omega$  at a temperature of  $20^\circ\text{C}$  and  $48\ \Omega$  at a temperature of  $60^\circ\text{C}$ .

The resistance of the thermistor does not change linearly with temperature.

Describe how the resistances at  $20^\circ\text{C}$ ,  $40^\circ\text{C}$  and  $60^\circ\text{C}$  show this is true.

.....  
.....  
.....  
.....  
.....[2]

32. M/J/16/22/Q9

Fig. 9.1 is a circuit diagram.

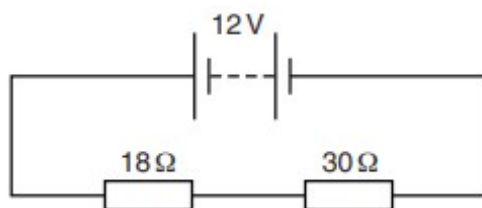


Fig. 9.1

The electromotive force (e.m.f.) of the battery is 12V.

(a) Explain what is meant by *electromotive force*.

.....  
 .....[2]

(b) An  $18\Omega$  resistor and a  $30\Omega$  resistor are connected in series with the battery.

(i) 1. State the relationship between the current  $I_B$  in the battery, the current  $I_1$  in the  $18\Omega$  resistor and the current  $I_2$  in the  $30\Omega$  resistor.

.....[1]

2. State an equation that relates the e.m.f.  $E$  of the battery, the potential difference  $V_1$  across the  $18\Omega$  resistor and the potential difference  $V_2$  across the  $30\Omega$  resistor.

.....[1]

(ii) Calculate the current in the battery.

current = .....[2]

(iii) Calculate the potential difference (p.d.) across the  $18\Omega$  resistor.

p.d. = .....[1]

- (iv) Calculate the power produced in the  $18\Omega$  resistor.

power = .....[2]

- (c) The resistors obey Ohm's law.

State Ohm's law and describe one limitation on this law.

.....  
.....  
.....[2]

- (d) The resistors are made from the same material, but have different cross-sectional areas and different lengths.

State the relationship between

- (i) the resistance  $R$  and the length  $l$  of a wire of constant cross-sectional area,

.....

- (ii) the resistance  $R$  and the cross-sectional area  $A$  of a wire of constant length.

.....  
[2]

33. M/J/16/21/Q8

Fig. 8.1 shows a circuit that contains a battery, a fixed resistor of resistance  $1000\ \Omega$  and a variable resistor  $R$ .

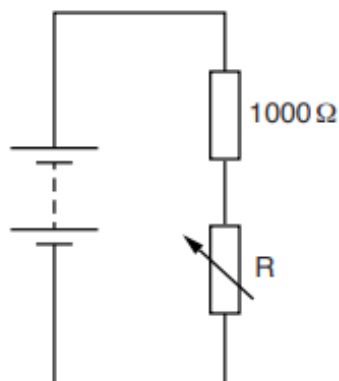


Fig. 8.1

- (a) (i) State the relationship between the current  $I_B$  in the battery, the current  $I_1$  in the fixed resistor and the current  $I_2$  in the variable resistor.

.....  
 .....[1]

- (ii) State an equation that relates the potential difference (p.d.)  $V_B$  across the battery, the p.d.  $V_1$  across the fixed resistor and the p.d.  $V_2$  across the variable resistor.

.....  
 .....[1]

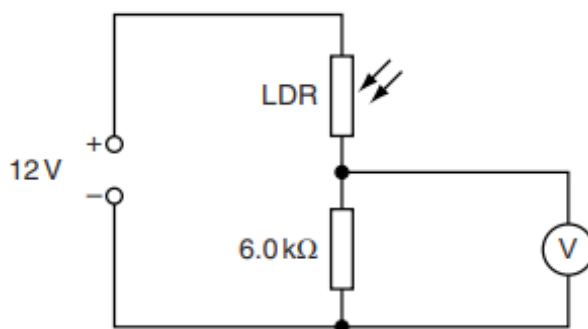
- (b) The p.d. across the battery is 9.0V. The resistance of  $R$  is set at  $500\ \Omega$ .

Calculate the current in  $R$ .

current = .....[2]

**34. O/N/15/21/Q8**

A potential divider is made from a light-dependent resistor (LDR) and a  $6.0\text{ k}\Omega$  fixed resistor. The potential divider is connected in series with a  $12\text{ V}$  d.c. power supply and a voltmeter is connected across the  $6.0\text{ k}\Omega$  resistor. Fig. 8.1 is the circuit diagram.



**Fig. 8.1**

A light shines on the LDR. The resistance of the LDR is  $2.0\text{ k}\Omega$ .

**(a)** Calculate

**(i)** the current in the LDR,

current = ..... [2]

**(ii)** the reading on the voltmeter.

reading = ..... [1]

**(b)** The brightness of the light on the LDR is gradually increased.

State and explain what happens to the reading on the voltmeter.

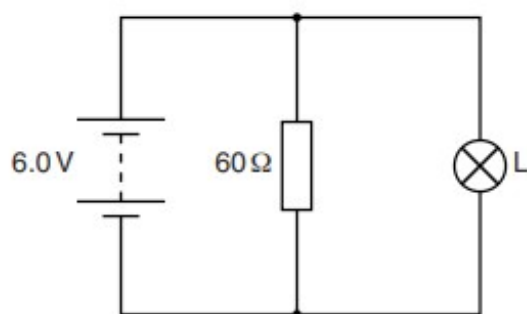
.....  
 .....  
 ..... [2]

**(c)** Suggest a use for a potential divider made from an LDR and a fixed resistor.

.....  
 ..... [1]

**35. M/J/15/21/Q6**

Fig. 6.1 shows a circuit that contains a resistor connected to a power supply of 6.0V and a lamp L.



**Fig. 6.1**

The resistor has a resistance of  $60\Omega$ . The lamp is marked 6.0V, 0.90W.

**(a)** Calculate

**(i)** the current in the resistor,

current = .....[2]

**(ii)** the current in the power supply.

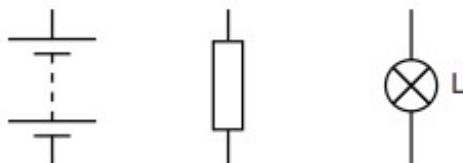
current = .....[2]

- (b) A second lamp is added to the circuit shown in Fig. 6.1.

The second lamp is in series with the  $60\Omega$  resistor, but is **not** in series with lamp L.

- (i) In the space below draw a circuit diagram of this new circuit.

The power supply,  $60\Omega$  resistor and lamp L have been drawn for you.



[1]

- (ii) The two lamps are identical.

Explain why the second lamp is dimmer than lamp L.

A calculation is not required.

.....  
.....  
.....[1]

36. O/N/14/22/Q7

The circuit of Fig. 7.1 includes an ammeter, a  $6.0\Omega$  resistor, a length of metal resistance wire and three  $1.5\text{V}$  cells connected in parallel.

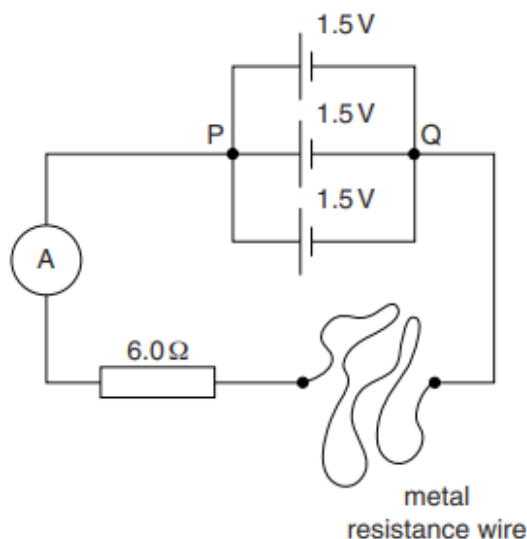


Fig. 7.1

(a) State

(i) one advantage of using cells in parallel rather than using a single cell,

.....  
 .....[1]

(ii) the potential difference (p.d.) between points P and Q in the circuit of Fig. 7.1.

p.d. = .....[1]

(b) (i) The ammeter in Fig. 7.1 reads  $0.075\text{A}$ .

Calculate the resistance of the resistance wire.

resistance = .....[3]

(ii) The temperature of the metal resistance wire increases.

State and explain the effect of this temperature increase on the ammeter reading.

.....  
 .....  
 .....[2]



37. O/N/14/21/Q11

A student makes a 2.0V battery by connecting two cells of electromotive force (e.m.f.) 2.0V in parallel. The battery, an ammeter with different ranges and three different resistors are used to set up the circuit shown in Fig. 11.1.

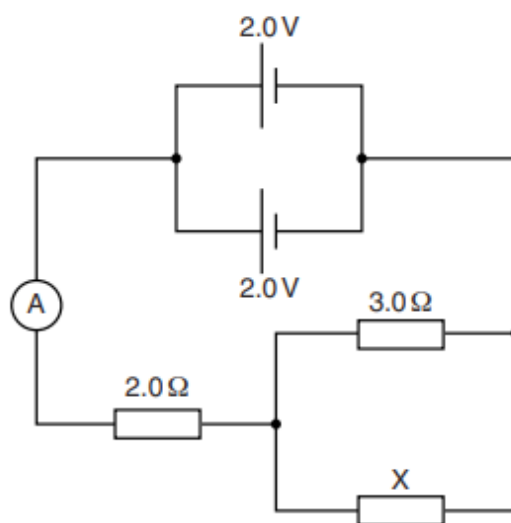


Fig. 11.1

- (a) (i) Explain what is meant by *electromotive force*.

.....  
 .....  
 .....[2]

- (ii) State one advantage of using two cells in parallel rather than using a single 2.0V cell.

.....  
 .....[1]

- (b) Resistor X and the 3.0Ω resistor have a combined resistance that is equal to 2.0Ω.

Calculate

- (i) the total resistance of the circuit,

total resistance = .....[1]

- (ii) the resistance of X.

resistance of X = .....[2]

- (c) (i) Determine the reading of the ammeter.

reading = .....[2]

- (ii) Suggest a suitable range for the ammeter.

.....[1]

- (d) The current in the  $2.0\Omega$  resistor is  $I_2$ .  
The current in the  $3.0\Omega$  resistor is  $I_3$ .  
The current in X is  $I_X$ .

State the equation that relates  $I_2$ ,  $I_3$  and  $I_X$ .

.....[1]

- (e) State the potential difference (p.d.) across

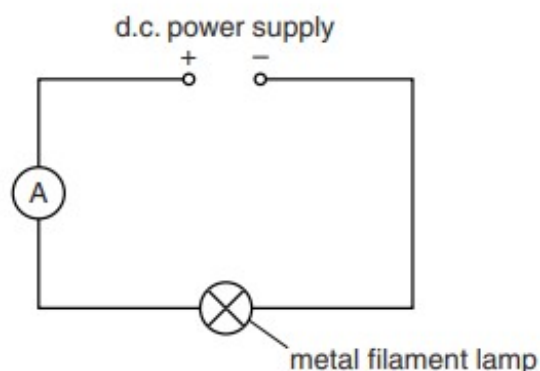
- (i) the  $2.0\Omega$  resistor,

p.d. = .....[1]

- (ii) the  $3.0\Omega$  resistor.

p.d. = .....[1]

- (f) The student sets up a second circuit using a variable d.c. power supply, an ammeter and a 12V metal filament lamp. The circuit is shown in Fig. 11.2.



**Fig. 11.2**

The d.c. power supply is set to 12V and the ammeter reading is 1.5 A.

The student changes the e.m.f. of the d.c. power supply to 6.0V. The lamp dims and the ammeter reading changes.

- (i) State and explain what happens to the resistance of the metal filament of the lamp.

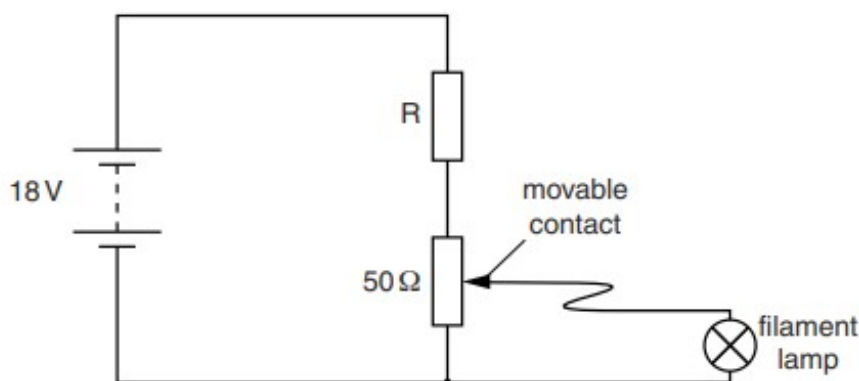
.....  
.....  
.....[2]

- (ii) State whether the new ammeter reading is less than, equal to or greater than 0.75 A.

.....[1]

**38. M/J/14/22/Q11**

A student sets up the circuit shown in Fig. 11.1.



**Fig. 11.1**

R is a fixed resistor in the circuit. The filament lamp is marked 12V, 0.25A.

The circuit is used to produce a current/voltage graph for the filament lamp. The ammeter and voltmeter needed are not shown.

To obtain different readings, the student changes the position of the movable contact.

**(a)** On Fig. 11.1, draw the symbols for an ammeter and a voltmeter in the correct positions. [3]

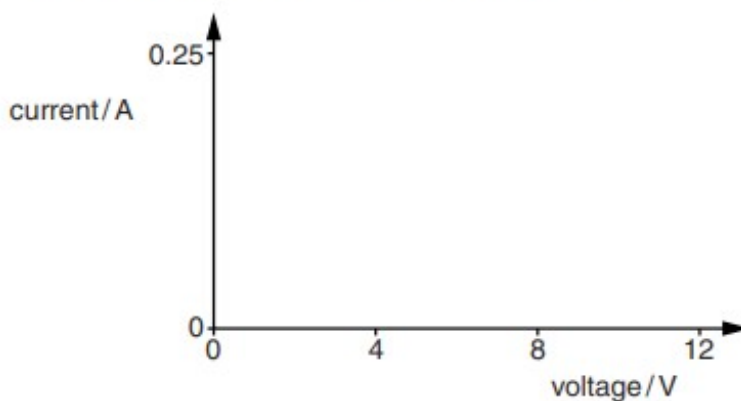
**(b)** Explain why it is sensible to include the resistor R in this circuit.

.....

.....

..... [2]

**(c) (i)** On Fig. 11.2, sketch a current/voltage graph for the lamp.



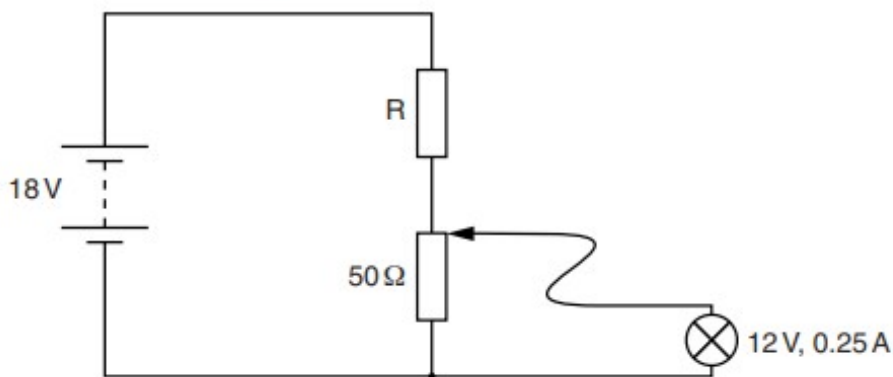
**Fig. 11.2**

[3]

- (ii) State and explain how a current/voltage graph for a fixed resistor is different from the graph for a filament lamp.

.....  
 .....  
 ..... [2]

- (d) Fig. 11.3 shows the position of the movable contact when the voltage across the lamp is 12 V and the current in the lamp is 0.25 A.



**Fig. 11.3**

Determine

- (i) the current in the  $50\ \Omega$  resistor,

current = ..... [2]

- (ii) the current in R,

current = ..... [1]

- (iii) the potential difference (p.d.) across R,

p.d. = ..... [1]

- (iv) the resistance of R.

resistance = ..... [1]

39. M/J/14/21/Q7

(a) Fig. 7.1 shows a variable potential divider (potentiometer) connected to a 6.0 V battery.

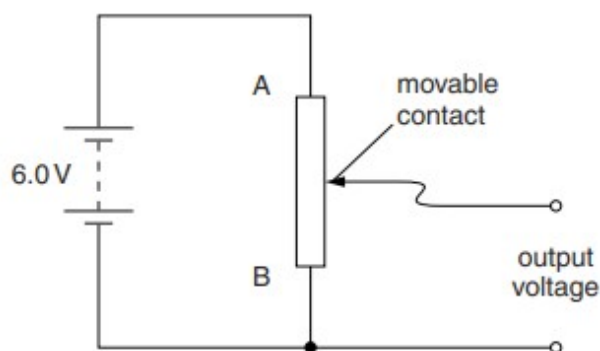


Fig. 7.1

The movable contact can slide between A and B. As it moves, the output voltage changes.

Complete the table below to show the output voltage when the contact is at A and at B.

| position of movable contact | A | B |
|-----------------------------|---|---|
| output voltage              |   |   |

[2]

(b) Fig. 7.2 shows a thermistor in a potential divider circuit that is used to monitor temperature. The fixed resistor has a resistance of  $2000\ \Omega$ .

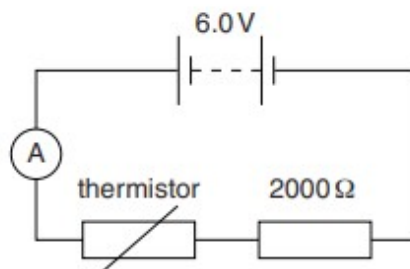


Fig. 7.2

(i) Explain why the voltage across the thermistor decreases as its temperature rises.

.....  
 ..... [2]

(ii) At one temperature, the thermistor has a resistance of  $1000\ \Omega$ . Calculate the current in the thermistor.

current = ..... [2]



40. M/J/14/21/Q11

- (a) A torch contains a filament lamp connected to a battery. The battery contains four 1.5V cells connected in series.

Fig. 11.1 shows the current/potential difference (voltage) graph for the filament lamp.

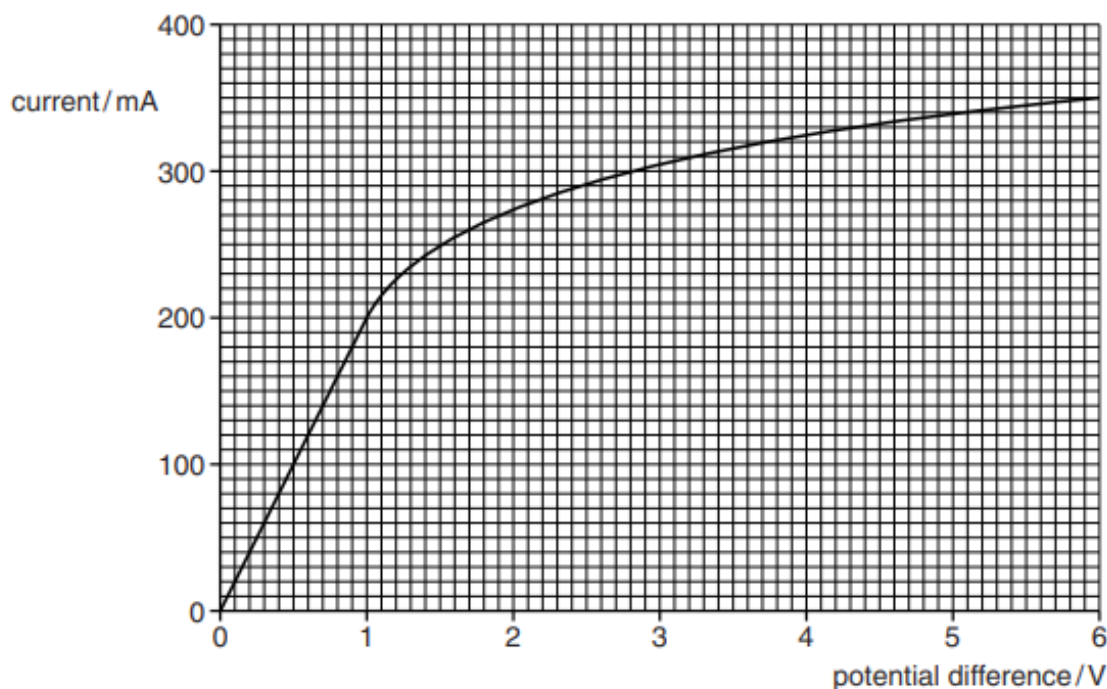


Fig. 11.1

- (i) State what is meant by the *potential difference* (p.d.) across the lamp.

.....  
 ..... [2]

- (ii) Fig. 11.1 shows that the current in the lamp is not proportional to the p.d.

1. State how Fig. 11.1 shows this.

.....  
 ..... [1]

2. Suggest why the current is not proportional to the p.d.

.....  
 ..... [1]

(iii) The lamp is at normal brightness and there is a p.d. of 6.0V across the lamp.

1. Use Fig. 11.1 to find the current in the lamp.

current = ..... [1]

2. Calculate the charge that passes through the lamp in 2.0 hours.

charge = ..... [2]

3. The battery in the torch is able to keep the lamp lit for 2.0 hours.  
Estimate the initial energy stored in the battery.

energy = ..... [2]

4. Explain why your value is only an estimate.

.....  
..... [1]

- (b) Suggest and explain an advantage of using two cells in parallel rather than a single cell.

.....  
.....  
..... [2]

- (c) Another torch contains a light-emitting diode (LED) instead of a filament lamp.  
The battery in the torch uses four 1.5V cells, some in series and some in parallel.  
The p.d. across the LED when in operation is 3.0V.

In the space below, draw a circuit diagram showing how the four cells, a switch and the LED are connected.

[3]



41. O/N/13/22/Q11

A metal filament lamp is connected to a power supply. The electromotive force (e.m.f.) produced by the supply can be varied. Fig. 11.1 is the circuit diagram.

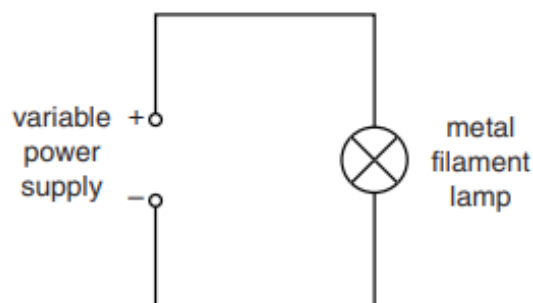


Fig. 11.1

(a) State what is meant by *electromotive force (e.m.f.)*.

.....

.....

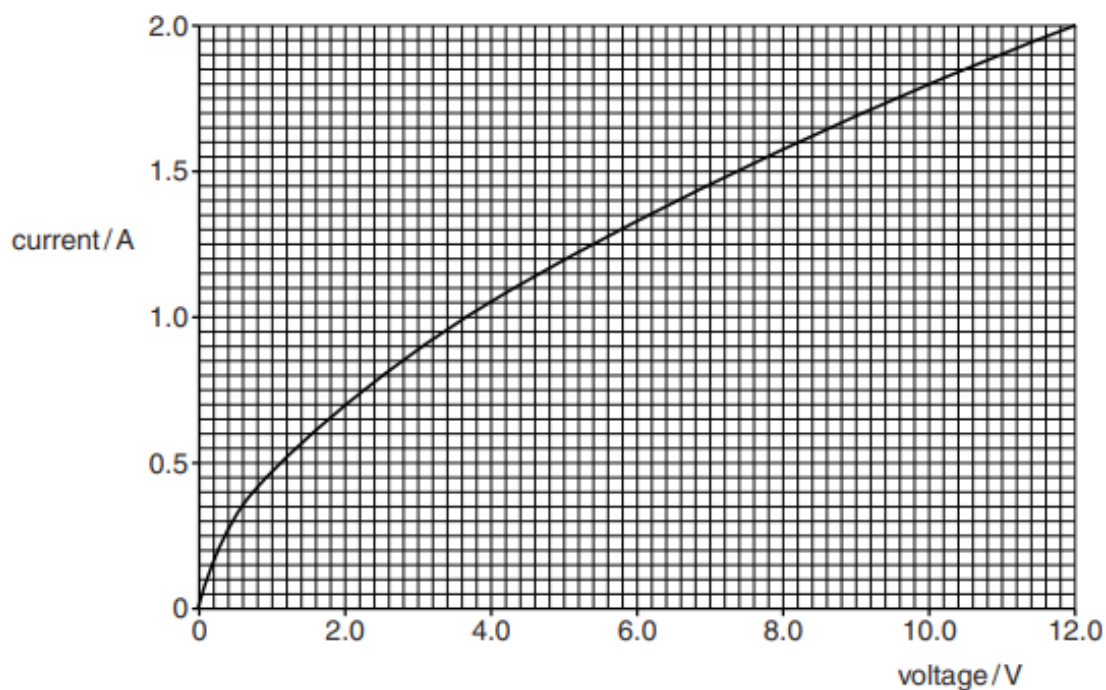
.....[2]

(b) Add appropriate circuit symbols to Fig. 11.1 to show the position of

(i) an ammeter that measures the current in the circuit, [1]

(ii) a voltmeter that measures the voltage across the lamp. [1]

(c) A student adds meters to the circuit and makes measurements of the current and voltage. He then plots the current/voltage graph shown in Fig. 11.2.



- (i) Using values from Fig. 11.2, calculate the resistance of the lamp when the current is 0.70 A. Give your answer to an appropriate number of significant figures.

resistance = ..... [3]

- (ii) State what, if anything, happens to the resistance of the lamp as the voltage increases.

.....

..... [1]

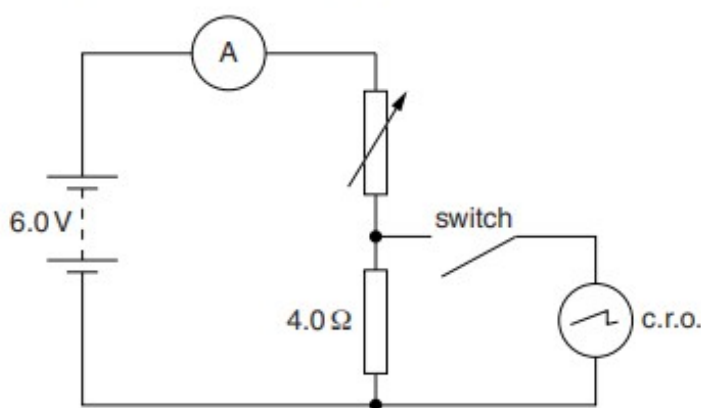
- (d) In normal use, the lamp is connected to a 12 V supply.

Use Fig. 11.2 to determine the power of the lamp when it is used in this way.

power = ..... [2]

**42. O/N/13/21/Q11**

An ammeter, a variable resistor and a  $4.0\Omega$  resistor are connected in series with a  $6.0\text{V}$  battery. A cathode-ray oscilloscope (c.r.o.) is connected in parallel with the  $4.0\Omega$  resistor through a switch. Fig. 11.1 is the circuit diagram.



**Fig. 11.1**

The switch is open.

- (a)** The variable resistor is adjusted so that it has a resistance of  $8.0\Omega$ .

Calculate

- (i)** the current measured by the ammeter,

current = .....[2]

- (ii)** the potential difference (p.d.) across the  $4.0\Omega$  resistor.

p.d. = .....[1]

- (b)** The resistance of the variable resistor is adjusted until the p.d. across the  $4.0\Omega$  resistor is  $5.0\text{V}$ .

State the effect that this adjustment has on

- (i)** the current measured by the ammeter,

.....[1]

- (ii)** the resistance of the variable resistor.

.....[1]

43. M/J/13/22/Q6

Fig. 6.1 shows the current-voltage graph for a filament lamp.

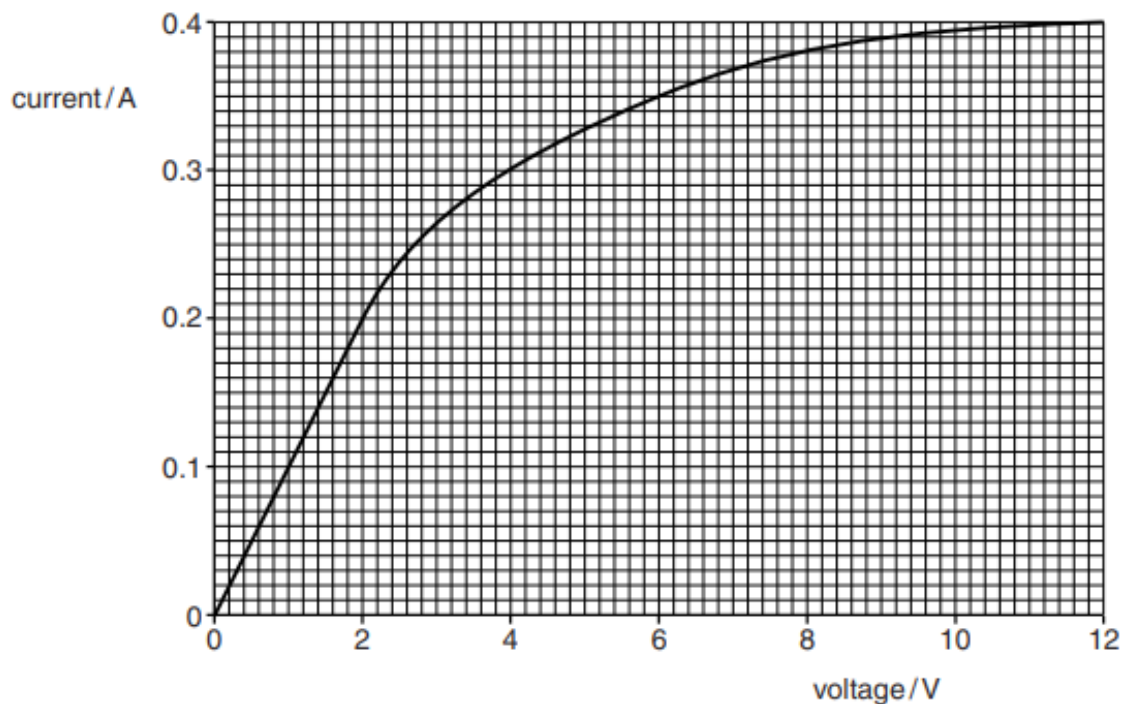


Fig. 6.1

- (a) (i) State the range of voltages where the resistance increases.

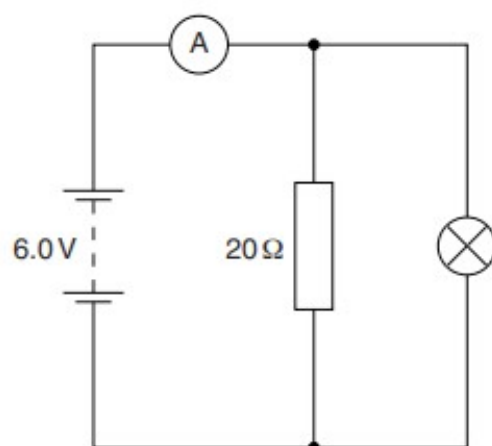
..... [1]

- (ii) State why the resistance of the lamp increases as the voltage increases.

.....

..... [1]

- (b) The filament lamp is connected in the circuit shown in Fig. 6.2.



**Fig. 6.2**

There are currents in the lamp, the  $20\Omega$  resistor and the ammeter.

- (i) State what is meant by an *electric current*.

.....  
..... [1]

- (ii) Use Fig. 6.1 to determine the current in the filament lamp.

current = ..... [1]

- (iii) Determine the current in the ammeter.

current = ..... [3]

44. M/J/13/21/Q6

Fig. 6.1 is a graph of current against potential difference (p.d.) for a length of metal wire.

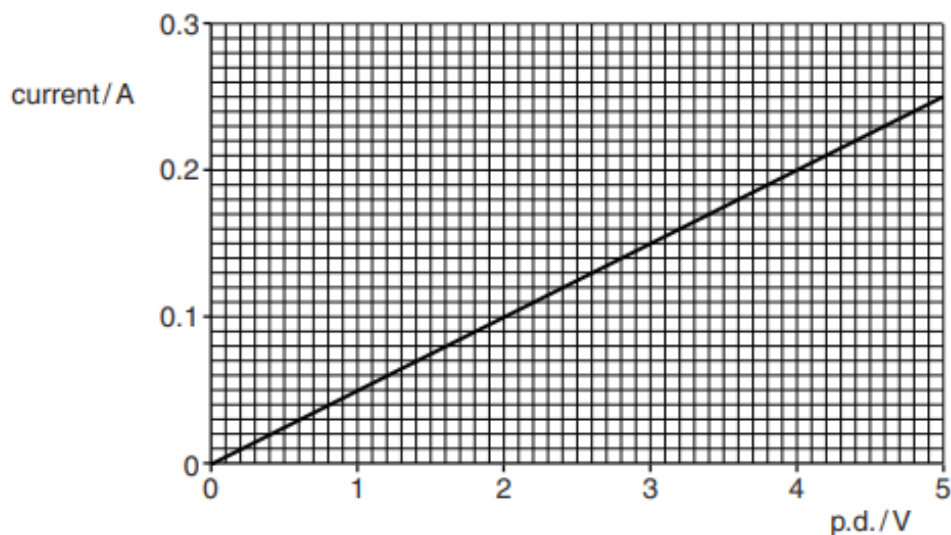


Fig. 6.1

- (a) The metal wire obeys Ohm's law. State Ohm's law in words.

.....  
 .....  
 ..... [2]

- (b) Calculate the resistance of the metal wire.

resistance = ..... [2]

- (c) A new wire is made from the same metal as the original wire. The two wires have the same length. The cross-sectional area of the new wire is half that of the original wire.

- (i) Calculate the resistance of the new wire.

resistance = ..... [1]

- (ii) On Fig. 6.1, draw a line to show how current varies with p.d. for the new wire. [2]

**45. M/J/13/21/Q7**

A student connects a thermistor in series with a power supply and a fixed resistor.

(a) In the space below, draw the circuit diagram of the apparatus.

[2]

(b) Values for potential difference (p.d.) and current are given in Fig. 7.1.

| p.d. across the thermistor/V | p.d. across the fixed resistor/V | e.m.f. of power supply/V | current in the thermistor/A | current in the fixed resistor/A |
|------------------------------|----------------------------------|--------------------------|-----------------------------|---------------------------------|
| 8.0                          | 4.0                              |                          | 0.018                       |                                 |

**Fig. 7.1**

The maximum power rating of the thermistor is 150mW.

(i) In the blank spaces in Fig. 7.1, insert the values of the electromotive force (e.m.f.) of the power supply and the current in the fixed resistor. [2]

(ii) Calculate the power produced in the thermistor.

power = ..... [2]

(iii) The e.m.f. of the power supply is doubled. Describe what happens to the thermistor.

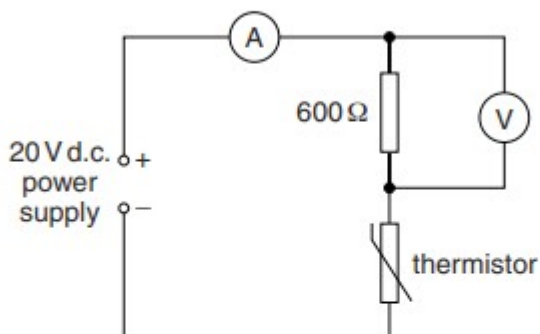
.....  
 ..... [1]



**46. O/N/12/21/Q8**

A  $600\ \Omega$  resistor and a thermistor are connected in series with an ammeter and a 20V d.c. (direct current) power supply. A voltmeter is in parallel with the resistor.

Fig. 8.1 is the circuit diagram.



**Fig. 8.1**

The ammeter reads 0.025 A.

**(a)** Calculate

**(i)** the reading on the voltmeter,

reading = ..... [2]

**(ii)** the resistance of the thermistor.

resistance = ..... [2]

**(b)** The temperature of the thermistor increases.

State what, if anything, happens to

**(i)** the resistance of the thermistor,

..... [1]

**(ii)** the ammeter reading and to the voltmeter reading.

ammeter reading: .....

voltmeter reading: .....

[1]

47. M/J/12/22/Q7

Fig. 7.1 shows two charged metal spheres.

One sphere has a positive charge and the other sphere has a negative charge.



Fig. 7.1

- (a) On Fig. 7.1, draw the electric field between the two spheres. [2]
- (b) The negative charge on the sphere is removed and a wire is used to connect the positive sphere to earth. The charge on the positive sphere decreases from  $4.8 \times 10^{-9} \text{ C}$  to zero in a time of  $2.0 \times 10^{-6} \text{ s}$ .

Calculate the average current in the wire.

current = .....[2]

**48. M/J/12/22/Q8**

A student investigates how the resistance of a wire depends upon its length. The student uses an ammeter, a voltmeter, a battery, a fixed resistor and the wire under test, all connected in an electrical circuit.

- (a) (i)** In the space below, draw a circuit diagram of the apparatus. Label the wire under test with the letter W.

[3]

- (ii)** Describe how the student obtains one complete set of results.

.....  
.....  
.....[2]

- (iii)** During the experiment, the student keeps the temperature of the wire constant.

- 1.** Suggest why it is sensible to keep the temperature of the wire constant.

.....  
.....[1]

- 2.** Suggest how the student keeps the temperature of the wire constant.

.....  
.....[1]

- (b) Fig. 8.1 shows part of a circuit containing three resistors X, Y and Z.

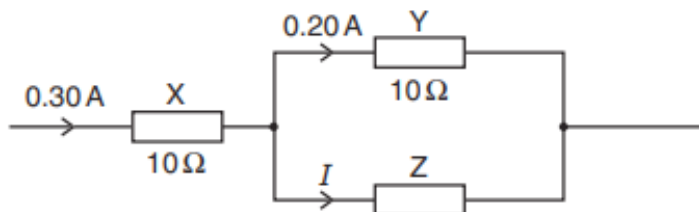


Fig. 8.1

The resistance of X and the resistance of Y are each  $10\ \Omega$ . The current through X is  $0.30\text{ A}$ . The current through Y is  $0.20\text{ A}$ .

- (i) Calculate the potential difference (p.d.) across Y.

p.d. = .....[2]

- (ii) Calculate the current  $I$  in resistor Z.

$I$  = .....[1]

- (iii) Explain in words, rather than by calculation, how Fig. 8.1 shows that the resistance of Z is larger than the resistance of Y. Use ideas about p.d. and current in your answer.

.....  
.....  
.....[2]

- (iv) Calculate the total resistance of the resistors X, Y and Z in this circuit.

resistance = .....[3]

49. M/J/12/21/Q7

Fig. 7.1 is the circuit diagram of a circuit set up by a student.

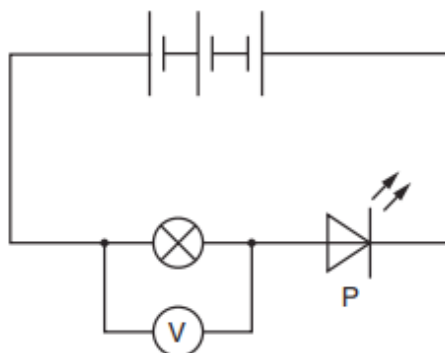


Fig. 7.1

The student uses three cells, each of electromotive force (e.m.f.) 1.5V. The cells are connected in series, as shown.

- (a) State the name of component P.

.....[1]

- (b) State what is meant by the *electromotive force* of a cell.

.....  
 .....[1]

- (c) The student expects the reading on the voltmeter to be 4.5V. The actual reading is 0.2V.

Suggest **two** reasons why the voltmeter reading is much less than 4.5V.

.....  
 .....  
 .....[2]

- (d) (i) In the space below, draw three cells connected in parallel.

[1]

- (ii) State one advantage of connecting cells in parallel.

.....  
 .....[1]

**50. O/N/11/22/Q6**

An electrical engineer measures the potential difference across a length of metal wire and the current in the wire. He does this for different values of the current.

- (a) (i) Draw a labelled circuit diagram of a circuit that enables the engineer to do this.

[3]

- (ii) Describe how the circuit is used.

.....

.....

.....

.....[2]

- (iii) State the equation that defines resistance.

.....[1]

- (b) The engineer performs the experiment and notices that the potential difference across the wire is directly proportional to the current in it. He calculates the resistance of the wire and plots a graph of the resistance against the current.

On Fig. 6.1, sketch the shape of this graph.

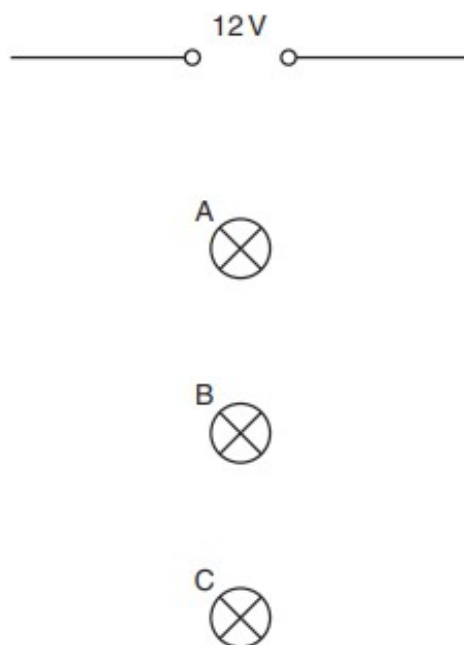


**Fig. 6.1**

[1]

**51. M/J/11/22/Q5**

Fig. 5.1 shows part of a low-voltage lighting circuit.



**Fig. 5.1**

The power supply voltage is 12V.

**(a)** On Fig. 5.1, complete the circuit, adding components as necessary, so that:

- the total current in the circuit can be measured,
- lamp A is on all the time,
- lamps B and C are in series with each other and are switched on or off together.

[3]

**(b)** All the lamps are on. The potential difference (p.d.) across lamp B is 8.0V and the current in lamp B is 50 mA.

**(i)** Calculate the resistance of lamp B.

resistance = ..... [2]

**(ii)** State the current in lamp C.

current = ..... [1]



**52. M/J/10/22/Q11**

- (a) Describe an experiment, including an ammeter and a voltmeter, to show how the resistance of a wire varies with temperature. Draw a circuit diagram.

.....

.....

.....

.....

.....

.....[4]

- (b) The filament in a lamp is made from metal wire.

- (i) Describe the variation with temperature of the resistance of the filament lamp.

.....

.....

.....[2]

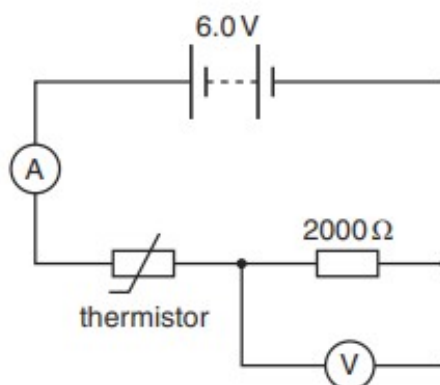
- (ii) On Fig. 11.1, sketch a graph of the variation of the current in the filament against the potential difference (p.d.) across it.



**Fig. 11.1**

[2]

(c) Fig. 11.2 is a circuit used to monitor changes in room temperature.



**Fig. 11.2**

A thermistor is connected in series with a 6.0V battery and a  $2000\Omega$  resistor.

(i) The temperature of the room increases. State and explain what happens to

1. the reading on the ammeter,

.....  
.....[2]

2. the reading on the voltmeter.

.....  
.....[2]

(ii) At a certain temperature, the reading on the voltmeter is 3.8V. Calculate the resistance of the thermistor at this temperature.

resistance = .....[3]

53. M/J/10/21/Q6

Fig. 6.1 shows a circuit used in an experiment to measure the current in a lamp for different values of the potential difference (p.d.) across the lamp.

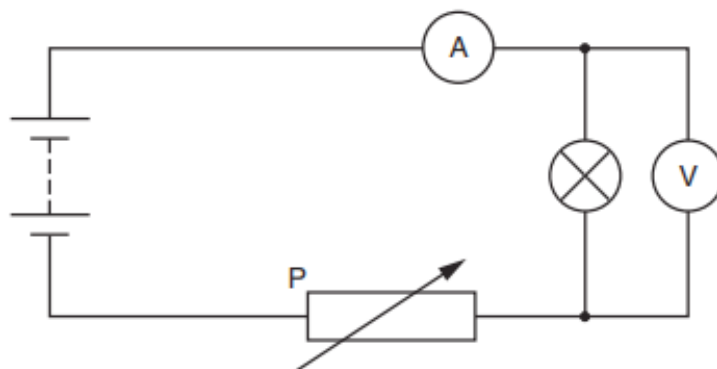


Fig. 6.1

(a) State the name of component P.

.....[1]

(b) On the axes below, sketch a graph of current against p.d. for a filament lamp.



[2]

(c) (i) State how the resistance of the lamp changes as the p.d. increases.

.....[1]

(ii) Explain how the graph in (b) shows the change you have described in (c)(i).

.....  
 .....  
 .....[1]

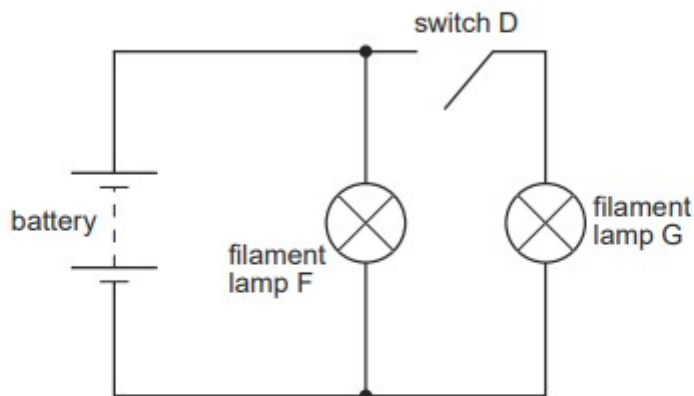
**54. O/N/08/Q7**

A motorcycle battery consists of six 2.0V cells in series. The battery supplies energy to the headlight.

- (a) State the total electromotive force (e.m.f.) of the battery.

..... [1]

- (b) The motorcycle headlight contains two identical filament lamps F and G. Filament lamp F is always lit but filament lamp G is turned on and off by switch D. Fig. 7.1 is the circuit diagram.



**Fig. 7.1**

When switch D is open, the battery supplies a current of 4.6 A.

Complete the table of Fig. 7.2.

|                    | current<br>supplied by<br>battery / A | current in<br>filament lamp<br>F / A | current in<br>filament lamp<br>G / A |
|--------------------|---------------------------------------|--------------------------------------|--------------------------------------|
| switch D<br>open   | 4.6                                   |                                      |                                      |
| switch D<br>closed |                                       |                                      |                                      |

**Fig. 7.2**

[3]

- (c) Calculate the energy supplied by the battery as an electric charge of 200C moves through the circuit.

energy supplied = ..... [2]

**55. M/J/08/Q6**

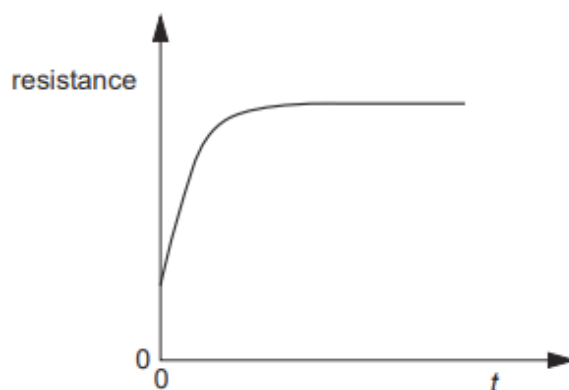
A filament lamp is labelled 240V, 0.20 A.

- (a) Calculate the resistance of the lamp at normal brightness.

resistance = ..... [2]

- (b) The lamp is connected to a 240V supply and switched on at time  $t = 0$ .

Fig. 6.1 shows the variation with  $t$  of the resistance of the lamp.



**Fig. 6.1**

Describe how the current in the lamp varies with  $t$ .

.....  
 .....  
 .....  
 .....  
 ..... [2]

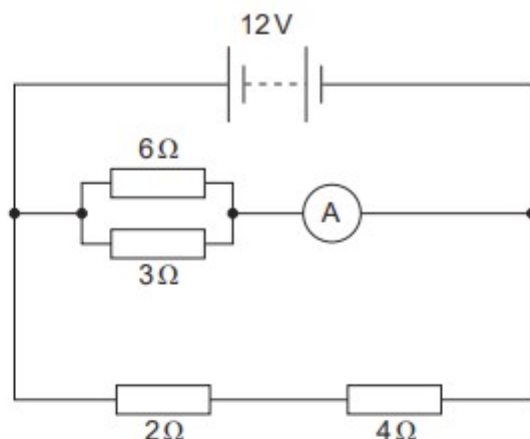
- (c) A second filament lamp has higher resistance.

State **two** differences between the two filaments that may cause the change in resistance.

- 1** The filament in the second lamp is .....  
 .....  
**2** The filament in the second lamp is .....  
 ..... [2]

**56. M/J/07/Q7**

Fig. 7.1 shows an electrical circuit containing a 12V power supply and a number of resistors.



**Fig. 7.1**

**(a)** Calculate the combined resistance of

**(i)** the  $2\Omega$  and  $4\Omega$  resistors in series,

resistance = ..... [1]

**(ii)** the  $3\Omega$  and  $6\Omega$  resistors in parallel.

resistance = ..... [2]

**(b)** Calculate the reading of the ammeter in Fig. 7.1.

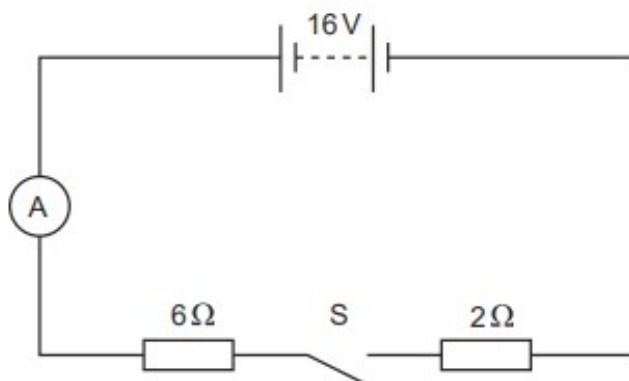
ammeter reading = ..... [2]

**(c)** Determine the potential difference across the  $4\Omega$  resistor.

p.d. = ..... [2]

**57. O/N/06/Q8**

Fig. 8.1 shows an electrical circuit using two resistors.



**Fig. 8.1**

- (a)** The switch S is open and the ammeter reading is zero.

State the value of the potential difference across the  $6\Omega$  resistor.

potential difference = ..... [1]

- (b)** Switch S is now closed.

- (i)** State the value of the total resistance of the circuit.

resistance = .....[1]

- (ii)** Calculate the current in the ammeter. State clearly the formula that you use.

current = .....[2]

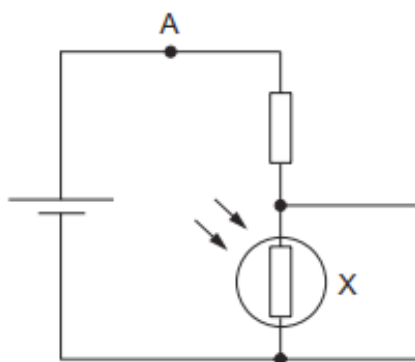
- (iii)** Calculate the potential difference across the  $6\Omega$  resistor.

p.d. = .....[2]



**58. M/J/05/Q7**

Fig. 7.1 shows an electrical circuit.



**Fig. 7.1**

- (a) On Fig. 7.1, draw an arrow at A to show the direction of flow of the electrons in the wire. [1]

- (b) What is the name of component X?

..... [1]

- (c) State and explain how the potential difference across X varies as the light shining on it becomes brighter.

.....

.....

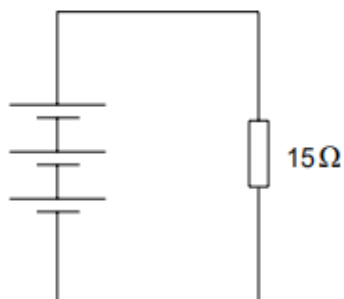
.....

.....

..... [2]

**59. M/J/05/Q8**

Three cells are connected in series making a battery, as shown in Fig. 8.1. The e.m.f. of each cell is 1.5 V. A resistance of  $15\ \Omega$  is connected to the battery.



**Fig. 8.1**

- (a) What is the total e.m.f. of the battery?

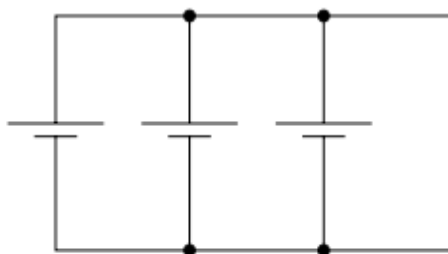
..... [1]

- (b) Calculate the current in the circuit.

State the equation that you use.

current = ..... [3]

- (c) A battery can be made from cells connected in parallel, as shown in Fig. 8.2.



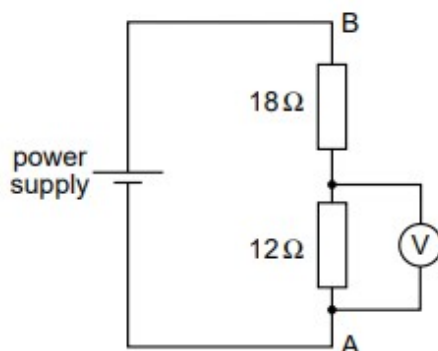
**Fig. 8.2**

State one advantage of connecting the cells in parallel.

.....  
 .....  
 .....  
 ..... [1]

**60. M/J/04/Q6**

Fig. 6.1 shows a circuit in which a voltmeter is placed across a resistor.



**Fig. 6.1**

The potential difference across the  $12\ \Omega$  resistor is  $4.0\ \text{V}$ .

The voltmeter has three different ranges:  $0$  to  $3.0\ \text{V}$ ,  $0$  to  $6.0\ \text{V}$  and  $0$  to  $30\ \text{V}$ . The best range for use in this circuit is  $0$  to  $6.0\ \text{V}$ .

**(a)** Explain why

**(i)** using the voltmeter on the range  $0$  to  $3.0\ \text{V}$  is unsuitable,

.....  
 .....

**(ii)** using the voltmeter on the range  $0$  to  $30\ \text{V}$  is unsuitable.

.....  
 .....

[2]

**(b)** **(i)** Calculate the current in the  $12\ \Omega$  resistor. State the formula that you use.

current = .....

**(ii)** Calculate the p.d. between A and B in Fig. 6.1.

p.d. = .....

[5]